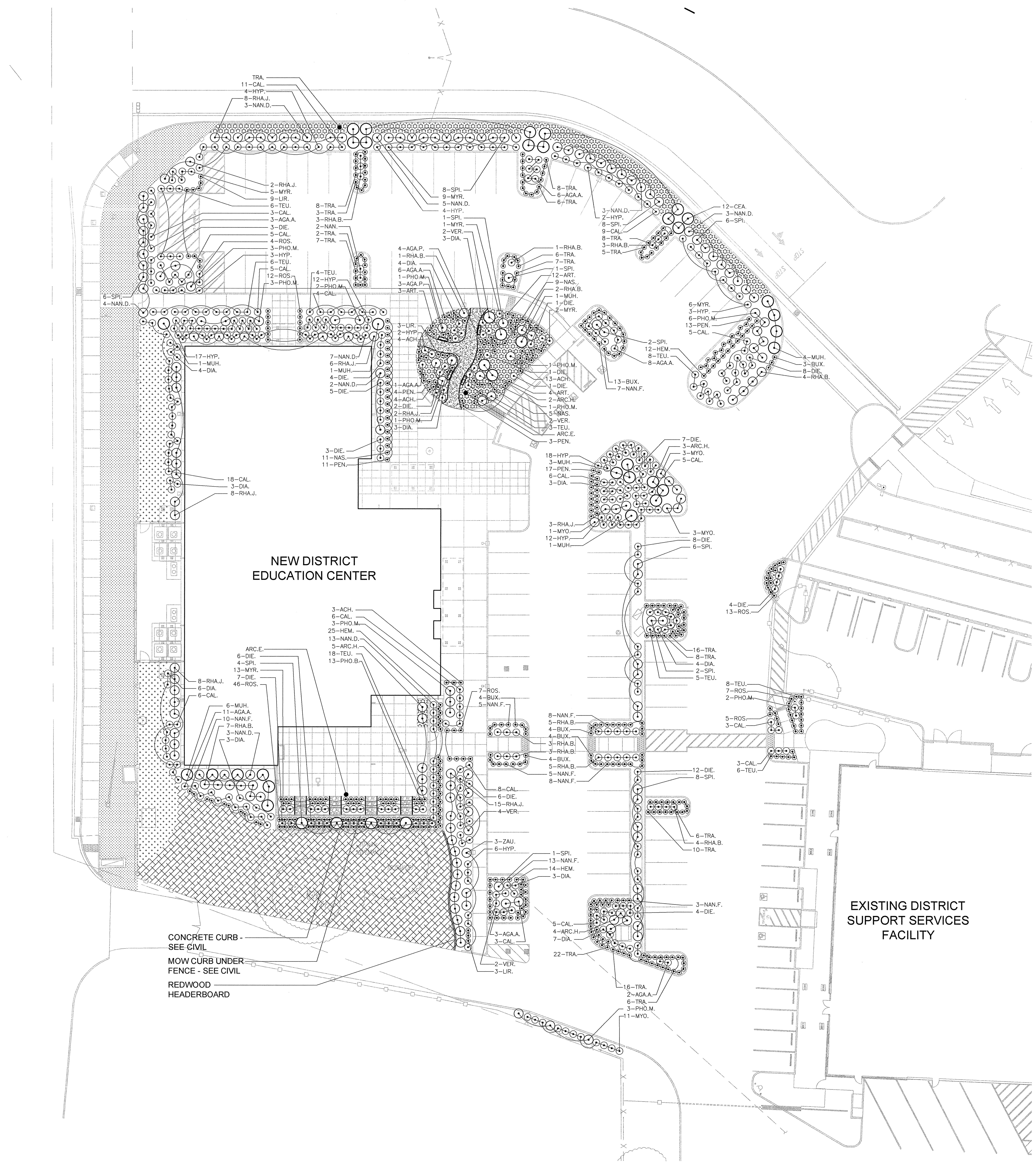
ONE INCH = TWENTY FEET
ONE-SIXTEENTH INCH = ONE FOOT
ONE-EIGHTH INCH = ONE FOOT
ONE-QUARTER INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE INCH = ONE FOOT
ONE AND ONE-HALF INCH = ONE FOOT



KEY	LANDSCAPE LEGEND
	SHRUBS
7 — AAA.	PLANT QUANTITY
	PLANT KEY
	LAWN - SOD
	NATIVE HYDROSEED MIX
	GROUNDCOVER
	BARK MULCH ONLY
	DECORATIVE GRAVEL PATH SEE DETAIL 5, SHEET L4.1
	REDWOOD HEADERBOARD SEE DETAIL 7, SHEET L4.1

PLANT MATERIAL LIST

WATER USE	SIZE	QUANTITY	KEY	BOTANICAL NAME ... COMMON NAME
				SHRUBS:
LOW	1 G.C.	40	AGA.A.	AGAPANTHUS AFRICANUS 'STORM CLOUD' ... LILY-OF-THE-NILE
LOW	1 G.C.	7	AGA.P.	AGAVE PYGMAEA 'DRAGON TOES' ... DRAGON TOES CENTURY PLANT
LOW	1 G.C.	24	ACH.	ACHILLEA 'NOVAACHDUS' ... MOON DUST YARROW
LOW	1 G.C.	14	ARC.H.	ARCTOSTAPHYLOS 'HOWARD MCMINN' ... HOWARD MCMINN MANZANITA
LOW	1 G.C.	19	ART.	ARTEMISIA SCHMIDTIANA 'SILVER MOUND' ... ANGEL'S HAIR
LOW	1 G.C.	32	BUX.	BUXUS MICROPHYLLA 'COMPACTA' ... BOXWOOD
LOW	1 G.C.	105	CAL.	CALLISTEMON VIMINALIS 'LITTLE JOHN' ... DWARF WEEPING BOTTLEBRUSH
LOW	5 G.C.	12	CEA.	CEANOTHUS THYRSIFLORUS 'SKYLARK' ... SKYLARK CEANOTHUS
LOW	1 G.C.	43	DIA.	DIANELLA REVOLUTA 'LITTLE REV' ... BLACK ANTHER FLAX LILY
LOW	1 G.C.	82	DIE.	DIETES VEGETA ... FORTNIGHT LILY
LOW	1 G.C.	51	HEM.	HEMEROCALLIS HYBRIDS 'STELLA D'ORO' ... DAYLILY
LOW	5 G.C.	83	HYP.	HYPERICUM MOSERIANUM ... GOLD FLOWER
LOW	1 G.C.	15	LIR.	LIRIOPE MUSCARI 'MAJESTIC' ... LILY TURF
LOW	1 G.C.	17	MUH.	MUHLENBERGIA RIGENS ... DEER GRASS
LOW	5 G.C.	36	MYR.	MYRTUS COMMUNIS 'COMPACTA' ... DWARF MYRTLE
LOW	1 G.C.	43	NAN.D.	NANDINA DOMESTICA ... HEAVENLY BAMBOO
LOW	1 G.C.	59	NAN.F.	NANDINA DOMESTICA 'FIRE POWER' ... FIRE POWER HEAVENLY BAMBOO
LOW	1 G.C.	25	NAS.	NASSELLA PULCHERA ... PURPLE NEEDLE GRASS
LOW	1 G.C.	48	PEN.	PENNISETUM MESSIACUM 'RED BUNNY TAILS' ... RED BUNNY TAILS
LOW	1 G.C.	13	PHO.B.	PHORMIUM 'TOM THUMB' ... TOM THUMB FLAX
LOW	1 G.C.	26	PHO.M.	PHORMIUM 'MAORI QUEEN' ... MAORI QUEEN FLAX
LOW	5 G.C.	41	RHA.B.	RHAPHIOLEPIS INDICA 'BALLERINA' ... DWARF INDIA HAWTHORN
LOW	5 G.C.	52	RHA.J.	RHAPHIOLEPIS INDICA 'JACK EVANS' ... PINK INDIA HAWTHORN
LOW	2 G.C.	94	ROS.	ROSA DRIFT RED ... RED DRIFT ROSE
LOW	5 G.C.	53	SPI.	SPIREA NIPPONICA 'SNOW MOUND' ... SPIREA
LOW	1 G.C.	64	TEU.	TEUCHRIUM CHAMAEDRYS ... GERMANDER
LOW	1 G.C.	10	VER.	VERBENA LILACINA 'DE LA MINA' ... LILAC VERBENA
LOW	1 G.C.	3	ZAU.	ZAUSCHNERIA CALIFORNICA ... HUMMINGBIRD TRUMPET
				GROUNDCOVER:
LOW	1 G.C.	18" O.C.	ARC.E.	ARCTOSTAPHYLOS 'EMERALD CARPET' ... EMERALD CARPET MANZANITA
LOW	1 G.C.	18	MYO.	MYOPORUM PARVIFOLIUM ... MYOPORUM
LOW	1 G.C.	144	TRA.	TRACHELOSPERUM JASMINOIDES ... STAR JASMINE

GENERAL LANDSCAPE REQUIREMENTS/NOTES

1. NO PLANTING SHALL BE STARTED UNTIL SPRINKLER IRRIGATION SYSTEM HAS BEEN TESTED BY CONTRACTOR IN PRESENCE OF OWNER'S REPRESENTATIVE AND NOTED DEFICIENCIES CORRECTED.
2. NO PLANTING SHALL BE STARTED UNTIL SOIL PREPARATION AND FINISH GRADING OPERATIONS HAVE BEEN COMPLETED AND APPROVED BY THE OWNER'S REPRESENTATIVE.
3. QUANTITIES SHOWN ON PLANT MATERIAL LIST ARE APPROXIMATE. PROVIDE QUANTITIES INDICATED ON LANDSCAPE PLAN.
4. PLANT MATERIAL IS SUBJECT TO APPROVAL OF OWNER'S REPRESENTATIVE
5. SEE SHEET L4.1 FOR PLANTING INSTALLATION DETAILS.

ENVIRONMENTAL REQUIREMENTS:

GENERAL: PROCEED WITH WORK IN ORDERLY AND TIMELY MANNER TO COMPLETE INSTALLATION OF LANDSCAPING WITHIN CONTRACT LIMITS.

PROTECTION:

EXISTING CONSTRUCTION: EXECUTE WORK IN AN ORDERLY AND CAREFUL MANNER TO PROTECT NEW CONCRETE WALKS, WORK OF OTHER TRADES, AND OTHER IMPROVEMENTS.

EXISTING UTILITIES: DETERMINE LOCATION OF UNDERGROUND UTILITIES AND PERFORM WORK IN A MANNER WHICH WILL AVOID POSSIBLE DAMAGE. HAND EXCAVATE, AS REQUIRED, TO MINIMIZE POSSIBILITY OF DAMAGE TO UNDERGROUND UTILITIES. MAINTAIN GRADE STAKES SET BY OTHERS UNTIL REMOVAL IS MUTUALLY AGREED UPON BY ALL PARTIES CONCERNED. BE RESPONSIBLE FOR PROTECTION OF EXISTING UTILITIES WITHIN CONSTRUCTION AREA; REPAIR DAMAGE TO UTILITIES THAT OCCUR AS A RESULT OF OPERATIONS OF THIS WORK.

LANDSCAPING: PROTECT LANDSCAPE WORK AND MATERIALS FROM DAMAGE DUE TO LANDSCAPE OPERATIONS, OPERATIONS BY OTHER CONTRACTORS AND TRADES AND TRESPASSERS. MAINTAIN PROTECTION DURING INSTALLATION AND MAINTENANCE PERIODS. TREAT, REPAIR OR REPLACE DAMAGED LANDSCAPE WORK AS DIRECTED AT NO ADDITIONAL COST TO CONTRACT.

ADVERSE CONDITIONS: WHEN CONDITIONS DETRIMENTAL TO SOD OR PLANT GROWTH ARE ENCOUNTERED, SUCH AS RUBBLE FILL, ADVERSE DRAINAGE CONDITIONS, OR OBSTRUCTIONS, NOTIFY OWNER'S REPRESENTATIVE BEFORE STARTING WORK.

PLANTING AND TURF INSTALLATION SEASONS AND CONDITIONS

NO WORK SHALL BE DONE WHEN GROUND IS FROZEN, SNOW COVERED, TOO WET OR IN AN OTHERWISE UNSUITABLE CONDITION FOR AMENDING SOIL, FINISH GRADING OR PLANTING.

SOIL TESTING/SOIL IMPROVEMENT:

SEE SPECIFICATIONS 32 90 00, SECTION 3.02 SOIL TESTING AND SECTION 3.03 PREPARATION.

SOIL PERCOLATION

EXCAVATE 10 PLANTING PITS IN RANDOM AREAS OF SITE. FILL EXCAVATED PLANTING PITS WITH WATER TO 1/2 DEPTH OF PIT. PITS SHOULD DRAIN WITHIN 4 HOURS. IF PLANTING PITS DO NOT DRAIN, NOTIFY INSPECTOR IMMEDIATELY. PLANTING SHALL NOT BE STARTED UNTIL OWNER'S REPRESENTATIVE HAS RESOLVED A METHOD TO REMEDY DRAINAGE ISSUE.

PLANT MATERIAL STANDARDS

PLANTS SHALL BE IN ACCORDANCE WITH AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) ANSI Z60.1–AMERICAN STANDARD FOR NURSERY STOCK, EXCEPT AS OTHERWISE STATED IN SPECIFICATIONS OR SHOWN ON DRAWINGS. WHERE DRAWINGS OR SPECIFICATIONS ARE IN CONFLICT WITH ANSI Z60.1, DRAWINGS AND SPECIFICATIONS SHALL PREVAIL. PRUNE, THIN OUT AND SHAPE TREES IN ACCORDANCE WITH ANSI STANDARD HORTICULTURAL PRACTICE. PRUNE TREES TO RETAIN REQUIRED HEIGHT AND SPREAD. UNLESS OTHERWISE DIRECTED BY LANDSCAPE ARCHITECT, DO NOT CUT TREE LEADERS, AND REMOVE ONLY INJURED OR DEAD BRANCHES FROM FLOWERING TREES.

EXISTING LANDSCAPE AND SPRINKLER IRRIGATION SYSTEM

WORK LIMITS OF THIS PROJECT EXTEND INTO AREAS THAT WERE PREVIOUSLY DEVELOPED UNDER OTHER CONTRACTS. PRIOR TO START OF WORK, CONTRACTOR SHALL MEET WITH OWNER'S REPRESENTATIVE TO LOCATE ALL CONNECTIONS CALLED FOR ON DRAWINGS. WORK LIMITS/FENCING SHALL BE LAID OUT BY CONTRACTOR AND VERIFIED BY OWNER'S REPRESENTATIVE. FENCE TO BE INSTALLED AND IRRIGATION SYSTEM SHALL BE TESTED WITH CONTRACTOR, INSPECTOR, AND OWNER'S REPRESENTATIVE PRESENT. DEFICIENCIES SHALL BE NOTED AT THIS TIME AND ARE THE RESPONSIBILITY OF OWNER. AT COMPLETION OF WORK, SYSTEM WILL AGAIN BE TESTED, DEFICIENCIES NOTED AT THIS TIME THAT WERE NOT NOTED PREVIOUSLY WILL BE RESPONSIBILITY OF CONTRACTOR. EXISTING LANDSCAPE THAT HAS BEEN DAMAGED DUE TO CONSTRUCTION SHALL BE RESTORED TO ORIGINAL CONDITION BY CONTRACTOR AT NO ADDITIONAL COST TO OWNER. PRIOR TO MAKING ANY CONNECTION TO MAIN LINE, CONTRACTOR SHALL NOTIFY OWNER 1 WEEK IN ADVANCE SO ADJUSTMENTS TO EXISTING WATERING PROGRAMS CAN BE MADE.

20' 10' 0 20' 40'
SCALE: 1"= 20'-0"

Dry Creek Joint Elementary School District
District Education Center

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Dry Creek Joint Elementary School District

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APPLICATION NO. 02-115456
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revision date: ______ by: ______

16-30

MTW group

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No. C-26867
RENEWAL DATE: 1/31/19
STATE OF CALIFORNIA

ARCHITECT

INITIAL BOX

NO.	DWG BY	DATE	REVIEWED

REVISIONS

NO.	DESCRIPTION	DATE	REV'D
	100% design developement	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017

JOB NO. Y1538.00

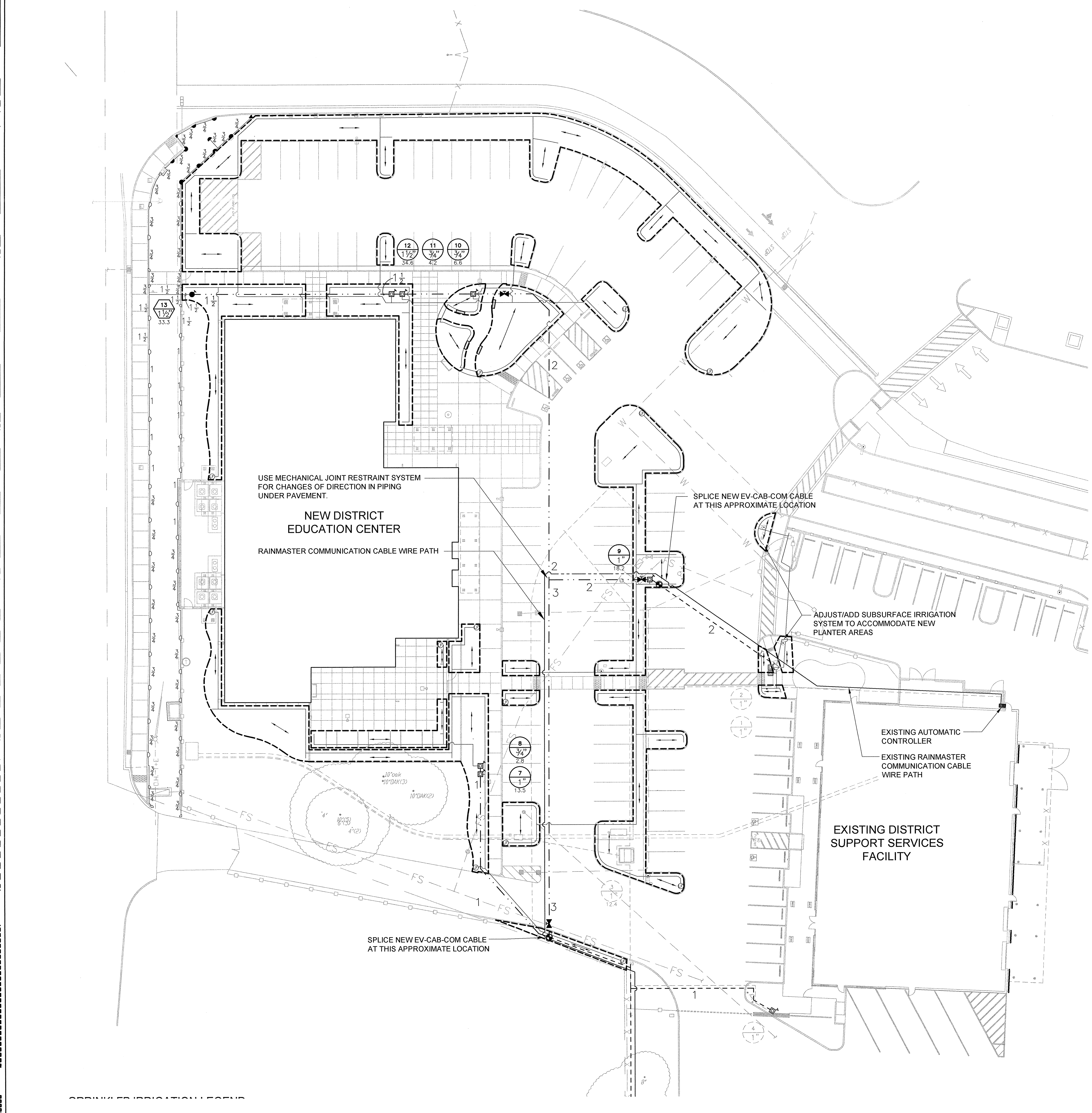
SHEET TITLE

Shrub And Groundcover Planting Plan

SHEET NO.

L2.1

SHEET 3 OF 7 TOTAL



KEY	SPRINKLER IRRIGATION LEGEND
	EXISTING AUTOMATIC CONTROLLER: RAINMASTER CENTRAL CONTROL SYSTEM: DX18-SWM/DX-HW/PMRKIT – 18 STATION CONTROLLER
	RAINMASTER COMMUNICATION CABLE WIRE PATH: RAINMASTER EV-CAB-COM IN 1" SCHEDULE 40 CONDUIT. INSTALL PULL BOXES EVERY 200', AT SPLICES, OR WHERE THE DIRECTION CHANGES.
	EXISTING RAINMASTER COMMUNICATION CABLE WIRE PATH:
	POINT OF CONNECTION: THIS SPRINKLER IRRIGATION SYSTEM IS DESIGNED TO OPERATE AT XX PSI STATIC WATER PRESSURE AND XXX PSI DYNAMIC WATER PRESSURE WITH A MAXIMUM FLOW OF 34.6 GPM. CONTRACTOR SHALL LOCATE EXISTING 4" IRRIGATION MAINLINE APPROXIMATELY AT THE LOCATION SHOWN. CONNECT AT THIS POINT AND EXTEND AS INDICATED ON DRAWINGS.
	GATE VALVE: TYPE: 3" SIZE AND SMALLER: NIBCO NO T-113 WITH A NON-RISING STEM AND HANDWHEEL. GATE VALVE INSTALLLED IN A VALVE BOX WITH TOP OF BOX SET FLUSH TO FINISH GRADE. GATE VALVE TO BE LINE SIZE.
	PRESSURE MAIN LINE: TYPE: 3" SIZE AND SMALLER: ASTM D1785, PVC SCH 40. TRENCH DEPTH: IN PLANTED AREAS: 24" MINIMUM COVER. UNDER PAVED AREAS: 36" MINIMUM COVER. PVC SCHEDULE 40 SLEEVES ARE REQUIRED FOR ALL PIPING UNDER PAVEMENT WHERE DISTANCE IS LESS THAN 16' IN LENGTH. PROVIDE LEEMCO MECHANICAL JOINT RESTRAINTS FOR CHANGES OF DIRECTION IN MAINLINE UNDER PAVEMENT.
	EXISTING PRESSURE MAIN LINE TO REMAIN:
	IRRIGATION LATERAL LINE: TYPE: PVC SCHEDULE 40, SOLVENT WELD ALL UNSIZED PIPE SHALL BE 3/4" SIZE. TRENCH DEPTH: IN PLANTED AREAS: POP-UP SPRAY HEADS – 12" MINIMUM COVER. UNDER PAVED AREAS: 24" MINIMUM COVER. PVC SCHEDULE 40 SLEEVES ARE REQUIRED FOR ALL PIPING UNDER PAVEMENT.
	LAWN POP-UP SPRAY HEADS: HUNTER: PROS-04 WITH CHECK VALVE AND MP 1000 NOZZLES. HALF, THIRD AND QUARTER SPRAY PATTERNS. (MAROON/LT.BLUE/OLIVE – 0.39 IN/HR)
	LAWN POP-UP SPRAY HEADS: HUNTER: PROS-04 WITH CHECK VALVE AND MP 2000 NOZZLES. HALF, THIRD AND QUARTER SPRAY PATTERNS. (BLACK/GREEN/RED – 0.39 IN/HR)
	LAWN POP-UP SPRAY HEADS: HUNTER: PROS-04 WITH CHECK VALVE AND MP CORNER NOZZLE. CORNER PATTERN. (TURQUOISE – N/A IN/HR)
	AUTOMATIC CONTROL VALVE: RAINBIRD PEB-PRS-D SERIES. VALVE SHALL HAVE PRESSURE REGULATION OPTION.
	AUTOMATIC DRIP IRRIGATION VALVE/FILTER/PRESSURE REGULATOR: RAINBIRD PEB SERIES WITH INLINE 30 PSI PRESSURE REGULATOR AND NETAFIM DISC FILTER.
LF LF	**SUBSURFACE INLINE DRIP SYSTEM:** NETAFIM TUBING TO BE TECHLINE CV AND IS TO BE INSTALLED IN THE DIRECTION OF THE ARROW AS SHOWN ON PLAN. TECHLINE TO BE INSTALLED 4" BELOW GRADE IN SHRUB BEDS. PVC SCHEDULE 40 LATERAL LINES SHOWN ON THE PLAN, NETAFIM TECHLINE CV NOT SHOWN. NETAFIM DRIP TUBING SPACING: SHRUB/GROUNDCOVER PLANTER AREAS: 18" ROW SPACING (0.6 GPH @ 18" SPACING) TECHLINE CV TUBING NOT TO EXCEED 400' IN A SINGLE RUN.
LF	**MANUAL LINE FLUSHING VALVE IN BOX:** MANUAL LINE FLUSHING VALVE TO BE INSTALL ON THE PVC PIPE IN AN EMITTER BOX BELOW GRADE.
00 / A1	INDICATES CONTROL VALVE AND STATION NUMBER
1" / 1"	INDICATES CONTROL VALVE SIZE
0.00 / 0.00	INDICATES GALLONS PER MINUTE

SPRINKLER IRRIGATION NOTES

1. COMPOSITE BASE SHEET: PROPOSED IMPROVEMENTS SHOWN ON DRAWINGS ARE SUPERIMPOSED ON A COMPOSITE BASE SHEET. THE COMPOSITE BASE SHEET IS A COMPILATION OF ARCHITECTURAL, ENGINEERING, AND OTHER DATA THAT IS PROVIDED. THE LANDSCAPE ARCHITECT SHALL NOT BE HELD LIABLE FOR CHANGES, INACCURACIES, OMISSIONS, OR ERRORS PERTAINING TO THE COMPOSITE BASE SHEET. CONTRACTOR SHALL BE RESPONSIBLE FOR REVIEWING THESE DOCUMENTS. ANY DISCREPANCIES NEED TO BE BROUGHT TO THE ATTENTION OF THE DESIGN TEAM AND RESOLVED PRIOR TO CONTINUATION OF WORK.
2. DESIGN PRESSURE SHOWN ON PLANS HAS BEEN FURNISHED BY WATER COMPANY OR WATER DISTRICT SERVING SITE. VERIFY PRESSURE ON-SITE PRIOR TO THE INSTALLATION OF ANY SPRINKLER IRRIGATION EQUIPMENT. IF THERE IS A DISCREPANCY, NOTIFY OWNER'S REPRESENTATIVE IMMEDIATELY IN WRITING SO ADJUSTMENTS CAN BE MADE BY LANDSCAPE ARCHITECT. FAILURE TO REPORT DISCREPANCIES AND CONTINUANCE OF WORK WILL RESULT IN ALL RE-DESIGN COSTS BEING CHARGED TO CONTRACTOR.
3. DETERMINE LOCATION OF UNDERGROUND UTILITIES. DAMAGE CAUSED BY INSTALLATION OF THIS WORK SHALL BE REPAIRED TO SATISFACTION OF GOVERNING AGENCY OR OWNER AT NO ADDITIONAL COST TO THE CONTRACT.
4. SPRINKLER OVER SPRAY SHALL NOT BE ALLOWED ON PUBLIC SIDEWALKS, BUILDING WALLS OR FENCES. MINIMUM OVERSPRAY MAY OCCUR IN PARKING AREAS. USE ADJUSTABLE NOZZLES WHENEVER POSSIBLE TO CONTROL SPRINKLER OVERSPRAY.
5. ALL LOCAL CODES AND ORDINANCES SHALL BE COMPLIED WITH. IF THERE IS A CONFLICT, NOTIFY OWNER'S REPRESENTATIVE IMMEDIATELY.
6. TESTING:
   A. PRESSURE TEST ALL UNDERGROUND PIPING AS FOLLOWS:
      MAIN LINE – AT STATIC PSI FOR 4 HOURS.
      LATERAL LINES – AT STATIC PSI FOR 2 HOURS.
   B. COVERAGE TEST:
      NOTE: PRIOR TO REQUESTING COVERAGE TEST, INSURE ALL HEADS ARE SET PLUMB, NOZZLES ARE ADJUSTED PROPERLY AND SYSTEM HAS BEEN CHECKED FOR AUTOMATION. REQUEST OWNER'S REPRESENTATIVES PRESENCE ON-SITE WHEN SPRINKLER SYSTEM IS COMPLETELY INSTALLED AND FULLY AUTOMATIC. PROVIDE ADEQUATE PERSONNEL AT THIS MEETING TO ADJUST AND FINE TUNE SYSTEM TO SATISFACTION OF OWNER'S REPRESENTATIVE.
7. LAYOUT ALL WORK PRIOR TO TRENCHING OPERATIONS TO DETERMINE IF MINOR MODIFICATIONS OR ADJUSTMENTS WILL BE REQUIRED.
8. INSTALL ALL SPRINKLER HEADS PERPENDICULAR TO SLOPES OR GRADE.
9. CONTROL WIRE SHALL BE UF-14, COLOR FOR LEAD AND WHITE FOR COMMON. SPLICES SHALL BE PERMITTED AT VALVE BOX LOCATIONS ONLY.
10. PROVIDE AND INSTALL AUTOMATIC CONTROLLER AND UF-14 CONTROL WIRE. ELECTRICAL SUBCONTRACTOR SHALL PROVIDE 110V SERVICE AND SERVICE HOOKUP FROM POWER SOURCE TO AUTOMATIC CONTROLLER.
11. COORDINATE ALL WORK WITH OTHER TRADES SO PROGRESS OF WORK IS NOT INTERRUPTED AND CAN BE COMPLETED IN A TIMELY MANNER.
12. NO PLANTING SHALL BE STARTED UNTIL ALL SPRINKLER WORK HAS BEEN TESTED AND APPROVED IN PRESENCE OF OWNER'S REPRESENTATIVE.
13. FOR SPRINKLER IRRIGATION INSTALLATION DETAILS, SEE SHEET NO. L4.1 & L4.2.

PRE-CONSTRUCTION SPRINKLER IRRIGATION NOTES

1. PRIOR TO START OF CONSTRUCTION CONTRACTOR REQUIRED TO CONTACT
   NAME, TITLE: SHAUN SCHONDELMEYER
   PHONE NUMBER: (916) 770-8883
   TO SET UP A MEETING ON SITE TO OPERATE THE EXISTING SPRINKLER IRRIGATION SYSTEM AND DISCUSS THE MODIFICATIONS THAT ARE TO BE MADE TO THE EXISTING SYSTEM TO ACCOMMODATE FOR THE NEW CONSTRUCTION.
2. CONTRACTOR TO OPERATE AND PROGRAM EXISTING SPRINKLER IRRIGATION SYSTEM THAT IS TO REMAIN IN ORDER TO PROVIDE WATER TO THE EXISTING LANDSCAPE TO REMAIN.
3. CONTRACTOR TO REMOVE ALL EXISTING PIPE AND SPRINKLER HEADS WHEN THEY ARE IN NEW PLANTING AREAS.
4. CONTRACTOR TO RESTORE AND REPAIR ANY EXISTING SPRINKLER IRRIGATION SYSTEM OR EXISTING LANDSCAPE WHICH IS IN AREAS TO REMAIN THAT IS DAMAGED BY NEW WORK.
5. ALL WORK TO EXISTING SPRINKLER IRRIGATION SYSTEM TO BE COMPLETED PRIOR TO SITE DEMOLITION.

N

20' 10' 0 20' 40'

SCALE: 1"= 20'-0"

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revision date: ____________ by: ____________

16-30

MTW group

LICENSED LANDSCAPE ARCHITECT BRYAN HOLLIS WALKER 5453 SIGNATURE 10/18/16 DATE 7/31/17 RENEWAL DATE STATE OF CALIFORNIA

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LICENSED ARCHITECT BRIAN J. MAYTUM No. C-26867 RENEWAL DATE: 1/31/19 STATE OF CALIFORNIA

ARCHITECT

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	DSA approval	02/23/17	

DATE February 23, 2017

JOB NO. Y1538.00

SHEET TITLE

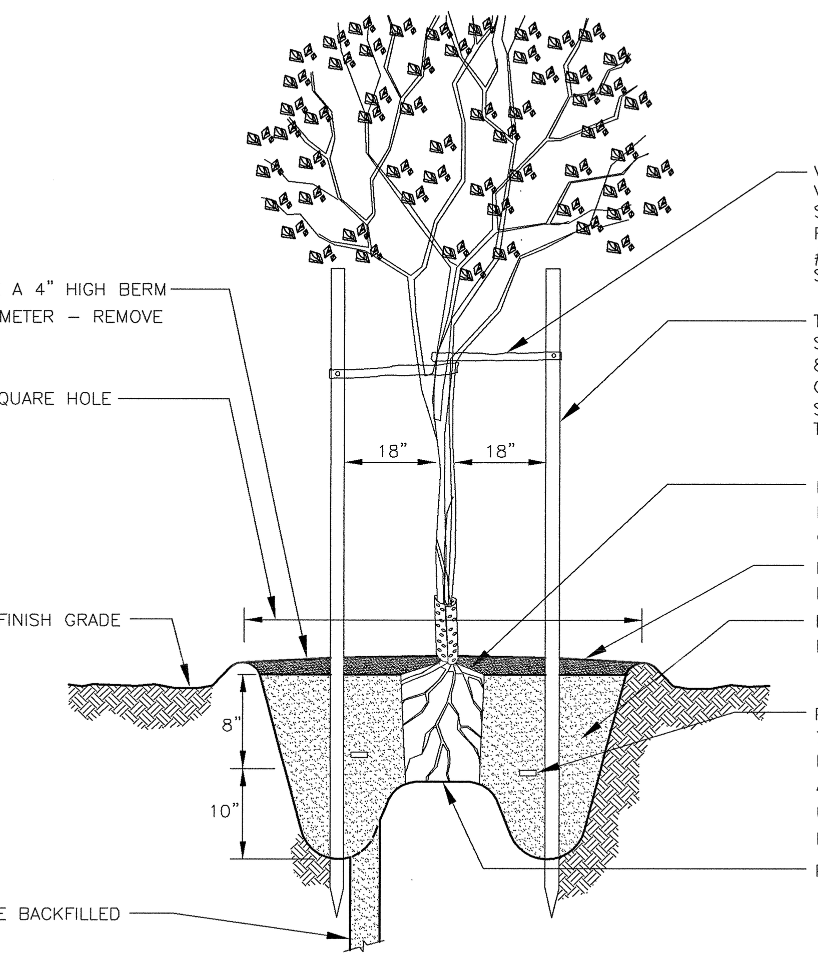
Sprinkler Irrigation Plan

SHEET NO.

L3.1

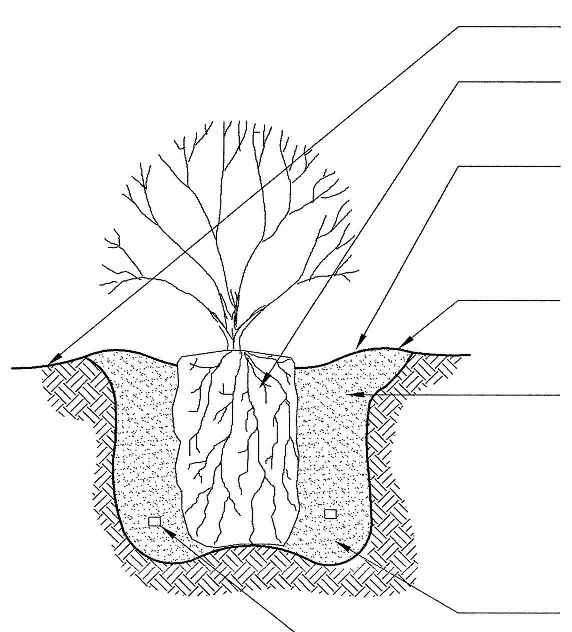
SHEET 4 OF 7 TOTAL

NOTES:
1. VITAMIN B-1: SUPERTHRIVE, LIQUINOX START OR EQUAL. APPLY AS PER MANUFACTURER'S INSTRUCTIONS WITH THE SECOND WATERING OF THE BASIN.
2. INSTALL ARBORGUARD TREE PROTECTION WHEN TREES ARE IN LAWN AREAS.
3. RETAIN ALL SMALL BRANCHES ON LOWER TRUNK.
4. KEEP NATURAL LEADER. DO NOT CUT OFF THE TOP OF THE TREE.
5. RELOCATE ALL TAGS AND LABELS FROM TRUNK TO A SIDE LIMB.
6. REMOVE NURSERY STAKE AT PLANTING TIME.
7. WATER THROUGHLY IN BASIN BY PLACING WATER HOSE NEAR TRUNK AND LETTING WATER TRICKLE SLOWLY.
8. TREE BRANCHES EXTENDING MORE THAN 4" OVER A WALKWAY MUST BE PRUNED UP TO AN 80" CLEARANCE PER CBC 1133B.8.6.1



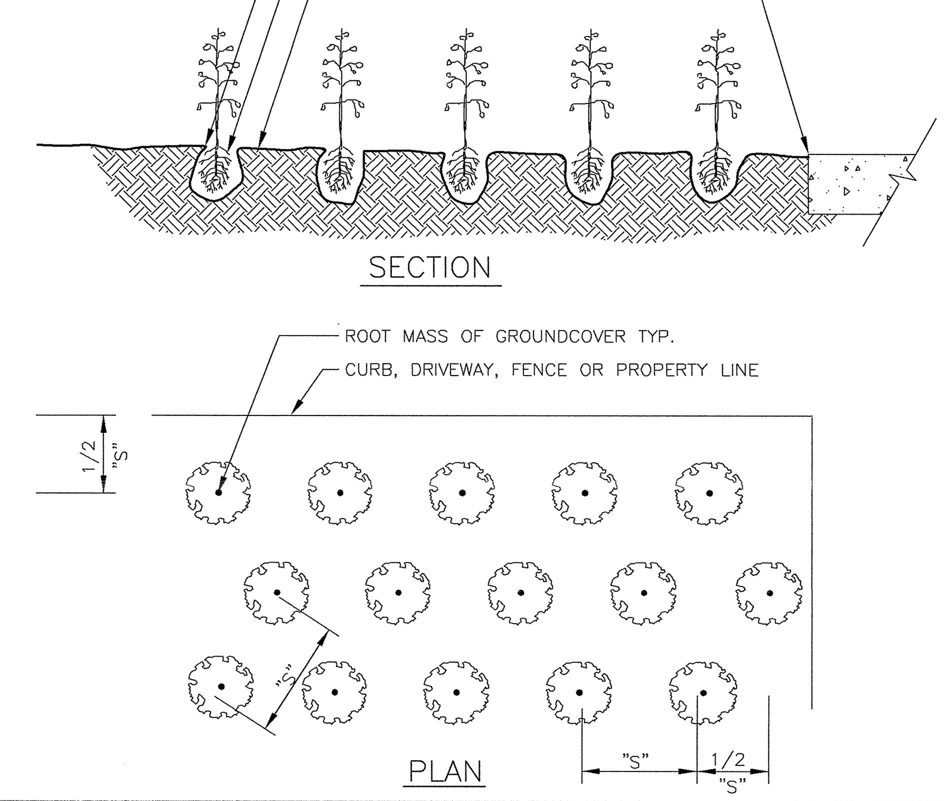
1 TREE PLANTING AND STAKING DETAIL

NOTE: VITAMIN B-1 SUPERTHRIVE LIQUINOX OR EQUAL. APPLY AS PER MANUFACTURE'S INSTRUCTIONS WITH SECOND WATERING OF THE BASIN.



2 SHRUB PLANTING DETAIL

NOTE:
1. "S" INDICATES SPACING SHOWN ON PLAN

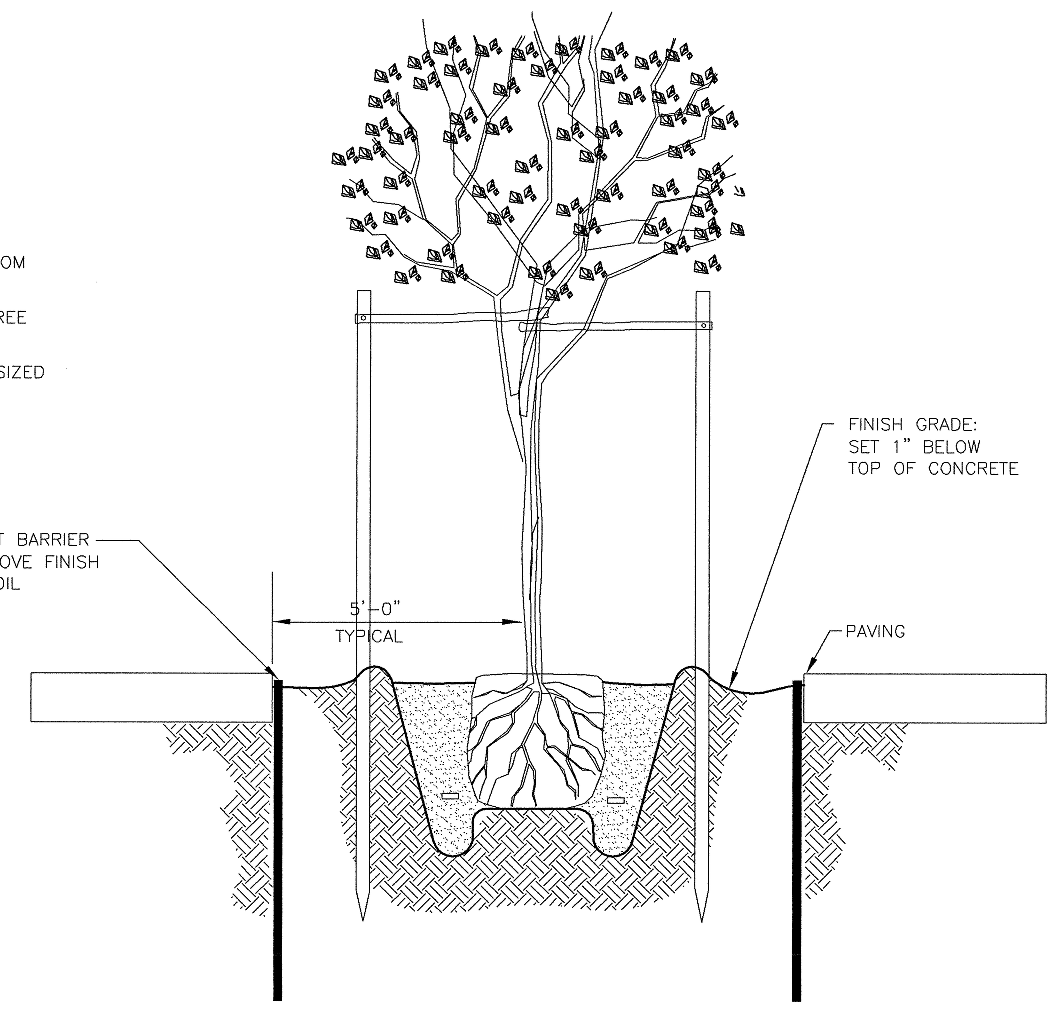


3 GROUNDCOVER/ANNUAL PLANTING DETAIL

NOTE:
1. SEE TREE PLANTING DETAIL FOR ADDITIONAL INFORMATION.

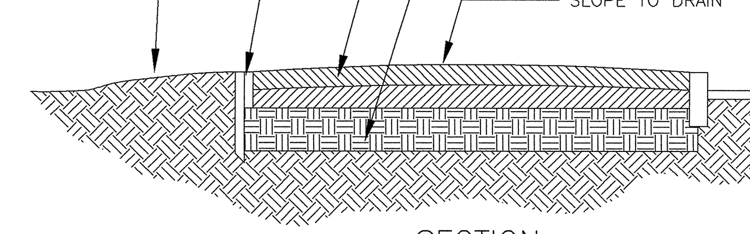
TREE ROOT BARRIER PROTECTION
1. TREE SHALL BE LOCATED IN CENTER OR MIDPOINT OF PLANTER.
2. TREE ROOT BARRIER IS TO BE INSTALLED ON EACH SIDE OF THE TREE WHERE THE DISTANCE FROM THE TREE TRUNK TO THE EDGE OF ANY HARD SURFACE PAVING IS FIVE FEET OR LESS.
3. TREE ROOT BARRIER SHALL BE INSTALLED ADJACENT TO EDGE OF PAVING AS DETAILED, USING TREE TRUNK AS CENTER OR MIDPOINT.
4. EACH LENGTH OF ROOT BARRIER USED SHALL BE TEN FEET LONG CENTERED ON THE TREE OR SIZED TO FIT PLANTER.
5. PLACE ROOT BARRIER SO THAT VERTICAL RIBS FACE TOWARD THE TREE.

ROOT BARRIER
BY: DEEP ROOT PARTNERS OR NDS INC. OR APPROVED EQ.
15 GAL SIZE TREE OR LARGER:
24" MODULAR ROOT BARRIER PANEL



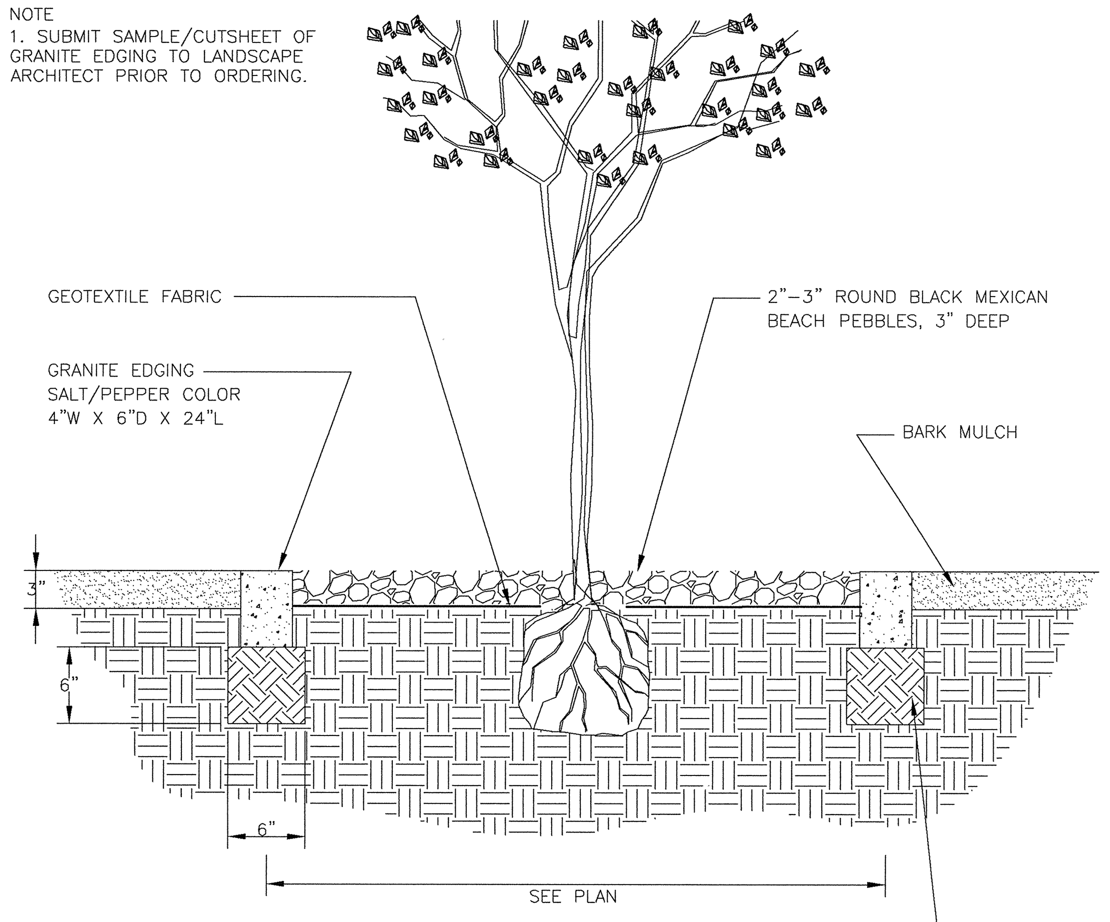
4 ROOT BARRIER CONTROL DETAIL

NOTES:
1. SUBMIT SAMPLE TO ARCHITECT FOR APPROVAL PRIOR TO ORDERING.
2. 1/4" MINUS AGGREGATE SHALL HAVE A MIXTURE OF FINE TO 1/4" SIZE PARTICLES WITH NO CLODS.
3. THE MATERIAL SHALL BE FREE OF VEGETATION, OTHER SOILS, DEBRIS AND ROCKS.
4. THE MATERIAL SHALL BE GREY IN COLOR.
5. THE MATERIAL SHALL BE OF SUCH NATURE THAT IT CAN BE COMPACTED READILY UNDER WATERING AND ROLLING.



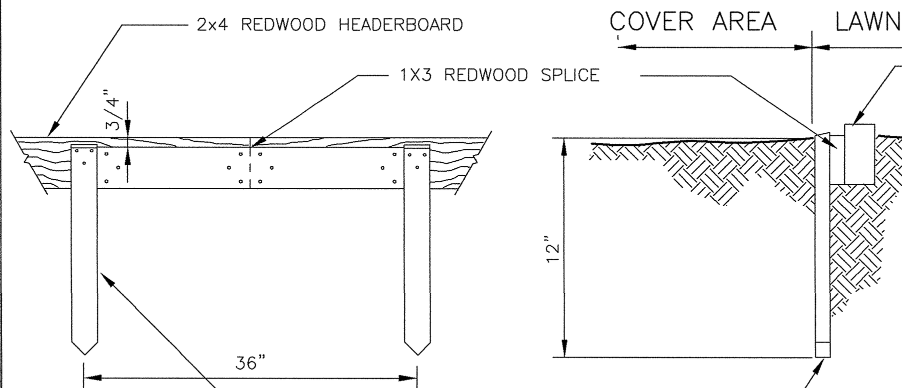
5 1/4" MINUS AGGREGATE/BINDER DETAIL

NOTE
1. SUBMIT SAMPLE/CUTSHEET OF GRANITE EDGING TO LANDSCAPE ARCHITECT PRIOR TO ORDERING.



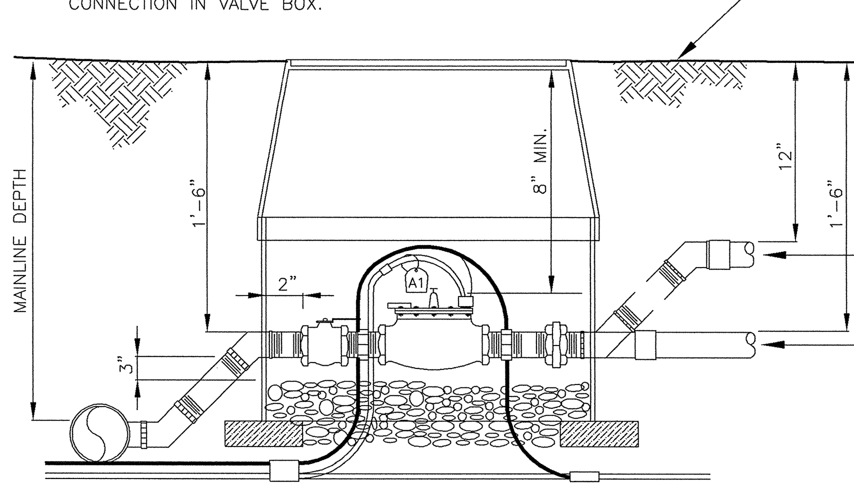
6 GRANITE EDGING DETAIL

NOTES:
1. WHEREVER POSSIBLE INSTALL STAKES AND SPLICES ON SHRUB/GROUNDCOVER SIDE.
2. ALL LUMBER TO BE REDWOOD CONSTRUCTION ROUGH (CALIFORNIA REDWOOD ASSOCIATION GRADING)
3. LAMINATE 1X4 BOARDS FOR CURVES. USE 1X3 REDWOOD AT SPLICES. STAGGER SPLICES.



7 REDWOOD HEADERBOARD DETAIL

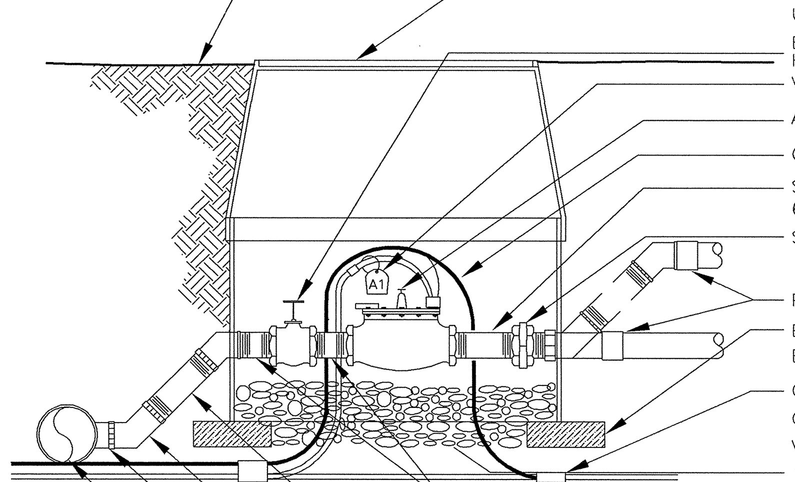
NOTES:
1. SET VALVE BOX LEVEL WITH FINISH GRADE.
2. INSTALL VALVE BOX EXTENSIONS AS REQUIRED.
3. CONNECT ALL WIRES WITH WEATHER TIGHT CONNECTORS. SEE SPECS. SPLICES IN WIRE SHALL BE PERMITTED ONLY AT VALVE LOCATIONS.
4. TAPE AND BUNDLE WIRE EVERY 10 FEET
5. PROVIDE 18" EXPANSION LOOP OF COILED WIRE AT EACH WIRE CONNECTION IN VALVE BOX.



6. FOR VALVES WITH PRESSURE REGULATION FEATURE, SET VALVE PRESSURE AS PER MANUFACTURERS RECOMMENDATIONS TO THE PRESSURE AS FOLLOWS:

POP-UP SPRAY HEADS:	30 PSI
INTERMEDIATE POP-UP ROTOR HEADS:	50 PSI
POP-UP ROTOR HEADS:	70 PSI
BUBBLER HEADS:	30 PSI

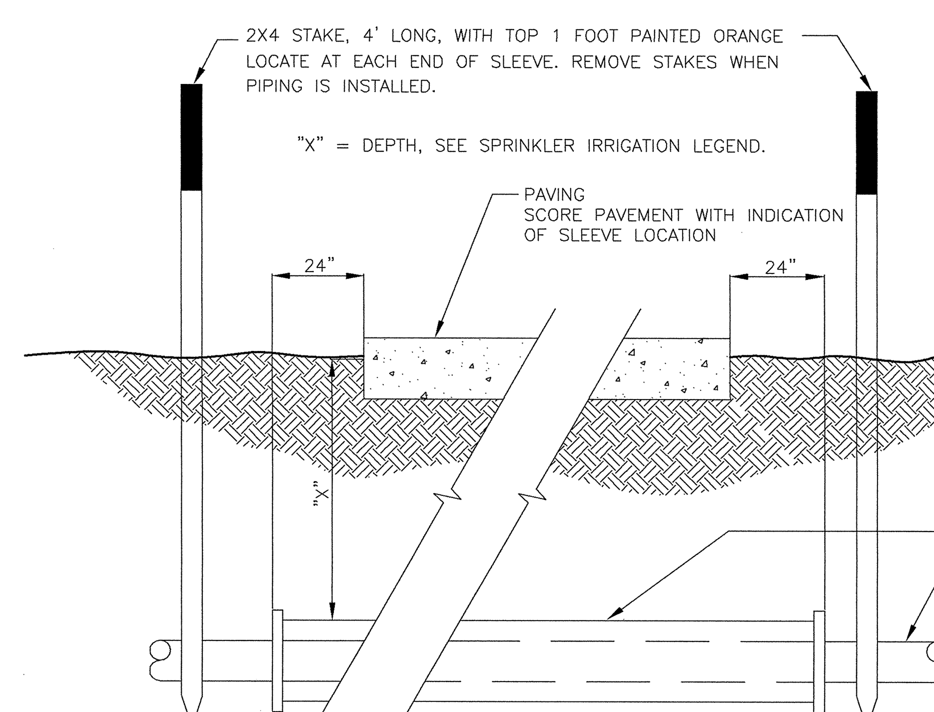
VALVE BOX, SETTINGS, AND CLEARANCES



COMPONENTS INSTALLATION

8 AUTOMATIC CONTROL VALVE/BALL VALVE DETAIL

NOTES:
1. ALL PIPE AND FITTINGS TO BE SCHEDULE 40, P.V.C. SEE PLAN FOR LOCATION.
2. SLEEVES TO BE LARGE ENOUGH TO ACCEPT THE PIPE AND FITTINGS TO BE ENCASED.
3. PROVIDE A SEPARATE SLEEVE FOR EACH LATERAL OR MAIN CROSSING.
4. PROVIDE A SEPARATE SLEEVE FOR CONTROL WIRE
5. TAPE ALL ENDS WITH DUCT TAPE TO PREVENT ENTRY OF SOIL.



9 SLEEVE DETAIL

Dry Creek Joint Elementary School District
District Education Center
8849 Cook Riolo Road, Roseville, CA 95747
Dry Creek Joint Elementary School District

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MTW group

CONSULTANT

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No. C-26867
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JOB NO. Y1538.00

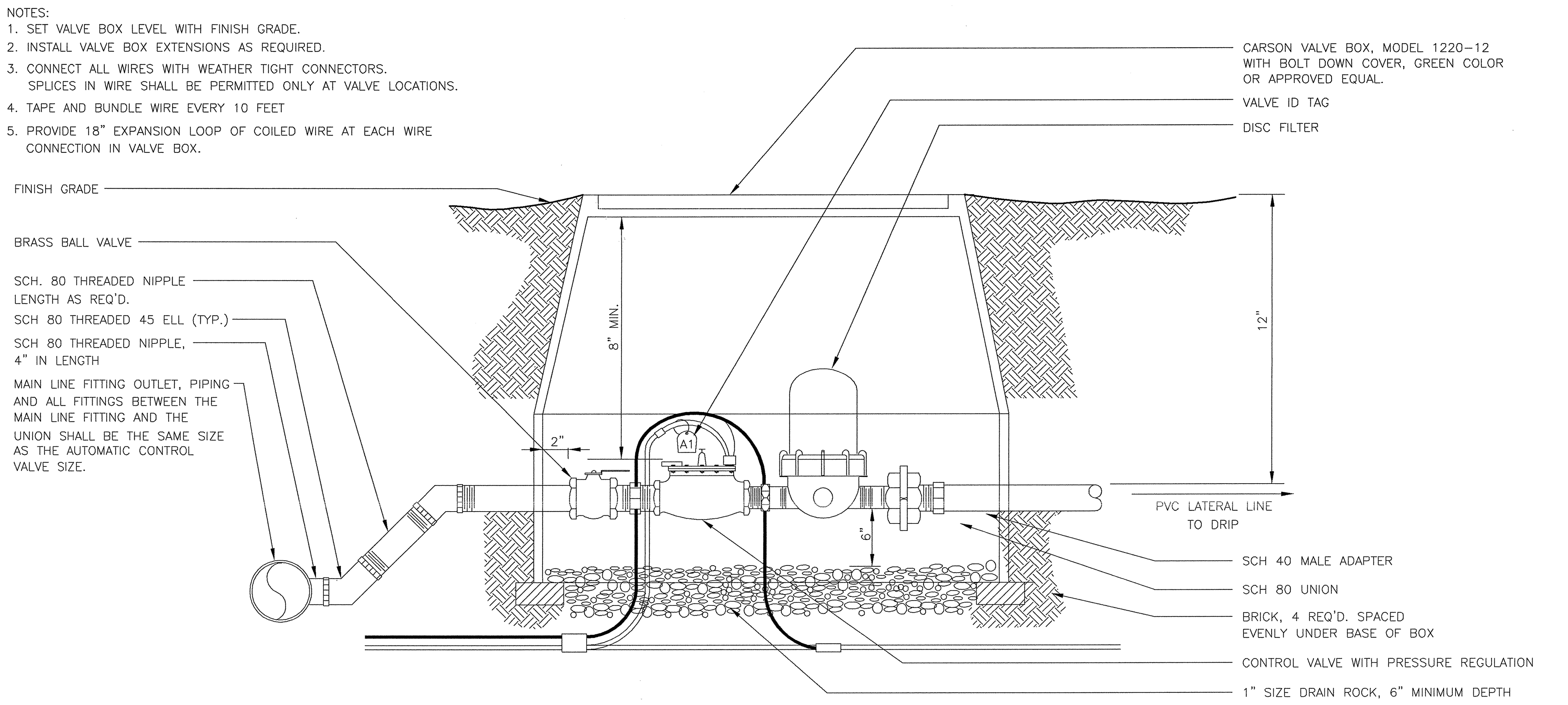
SHEET TITLE

Landscape Planting Details

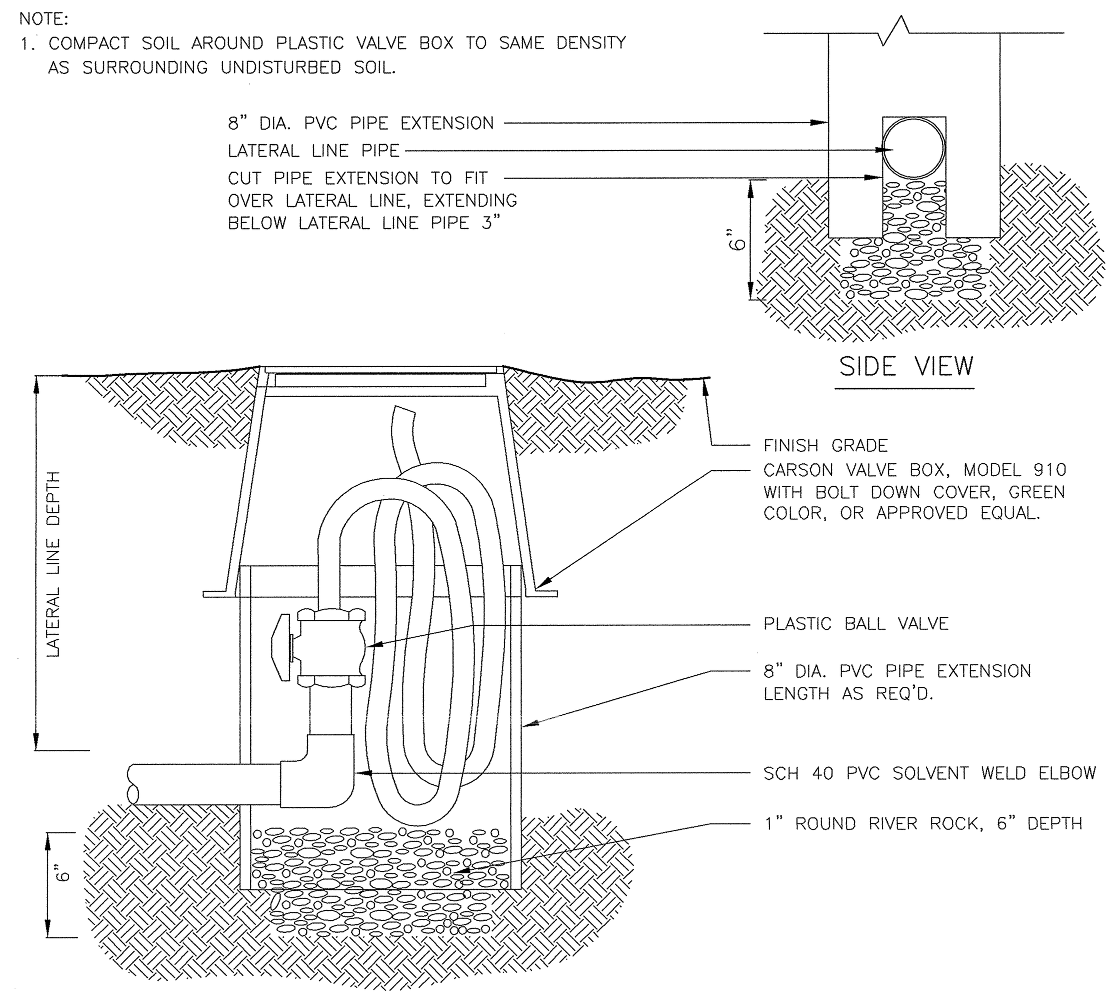
SHEET NO.

L4.1

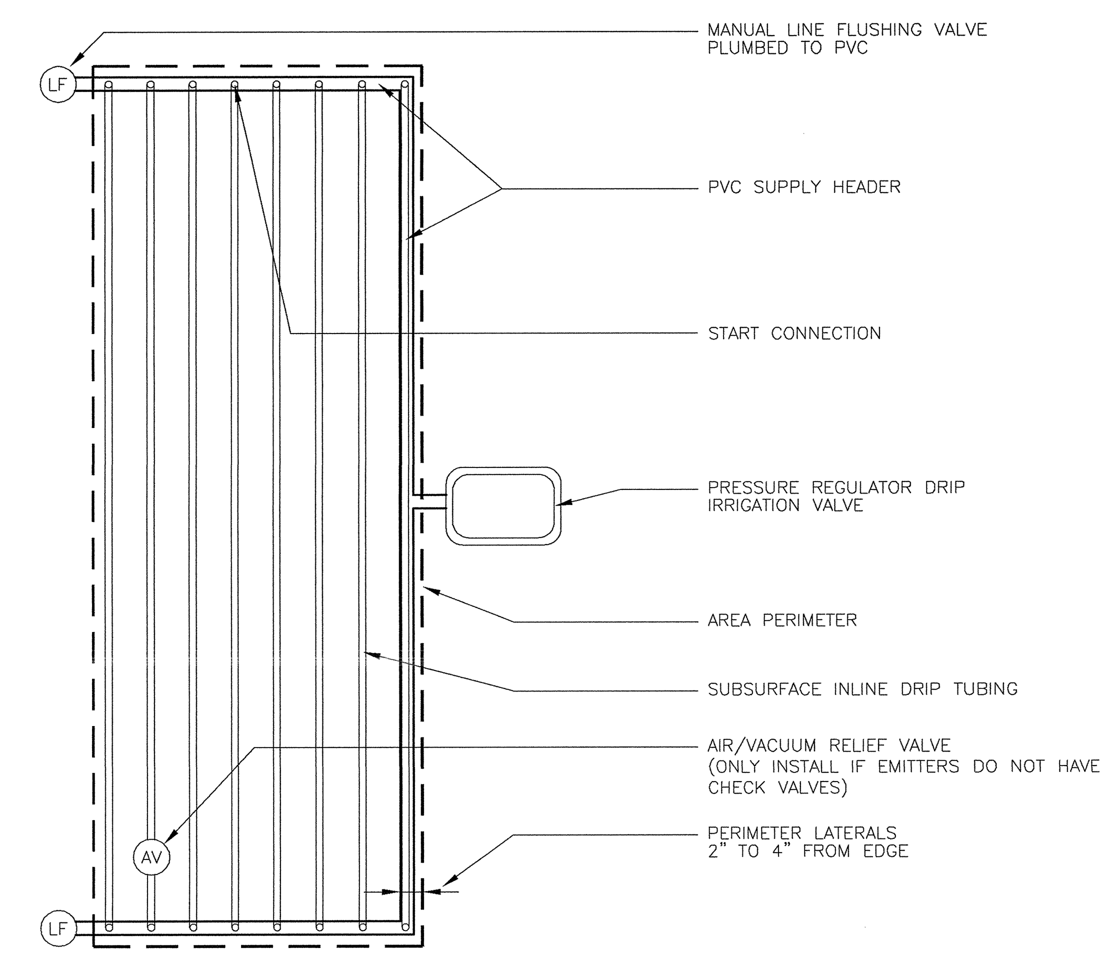
SHEET 5 OF 7 TOTAL



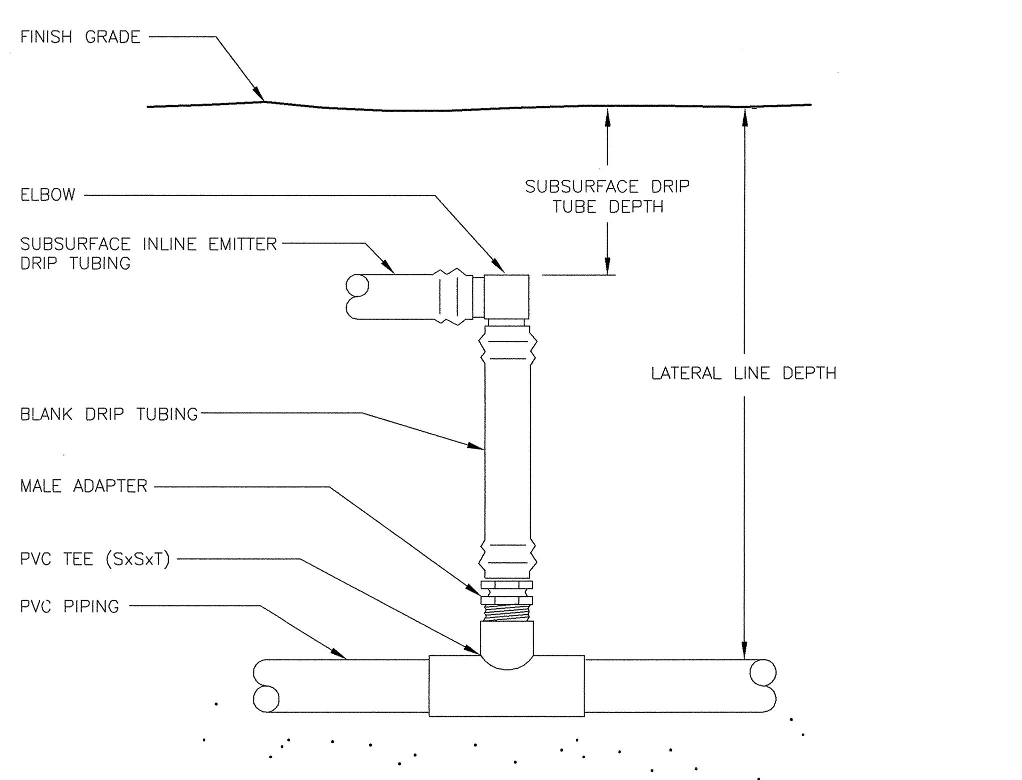
1 PRESSURE REGULATOR DRIP IRRIGATION VALVE DETAIL



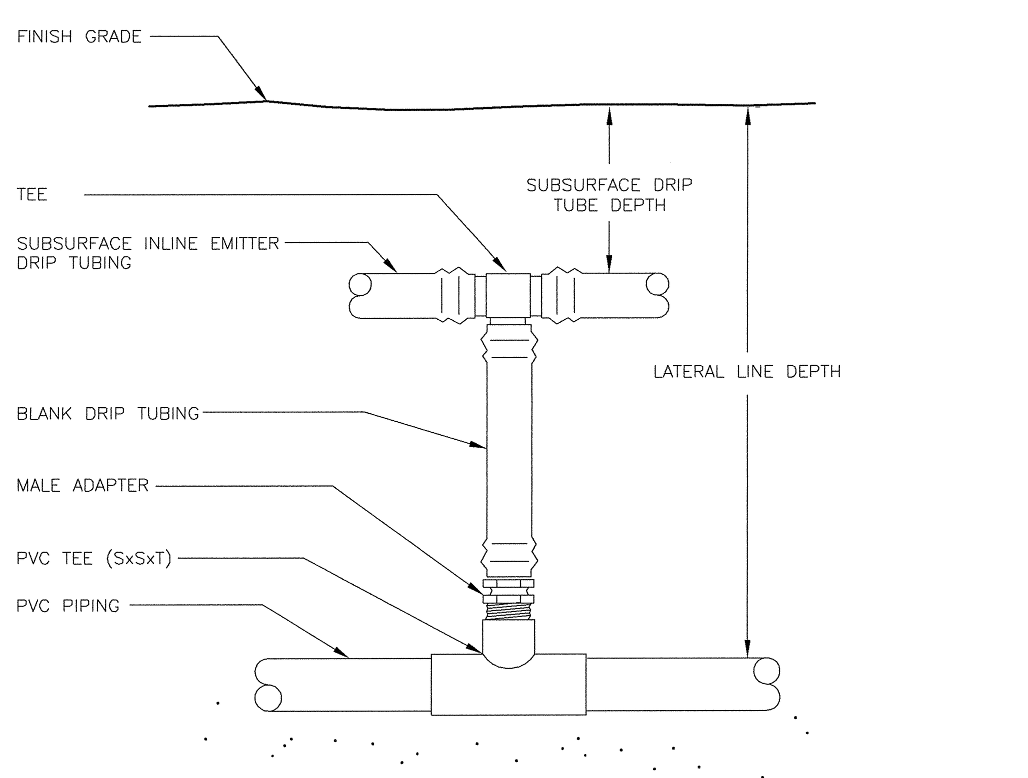
2 MANUAL FLUSH VALVE DETAIL



3 SUBSURFACE INLINE DRIP LAYOUT DETAIL



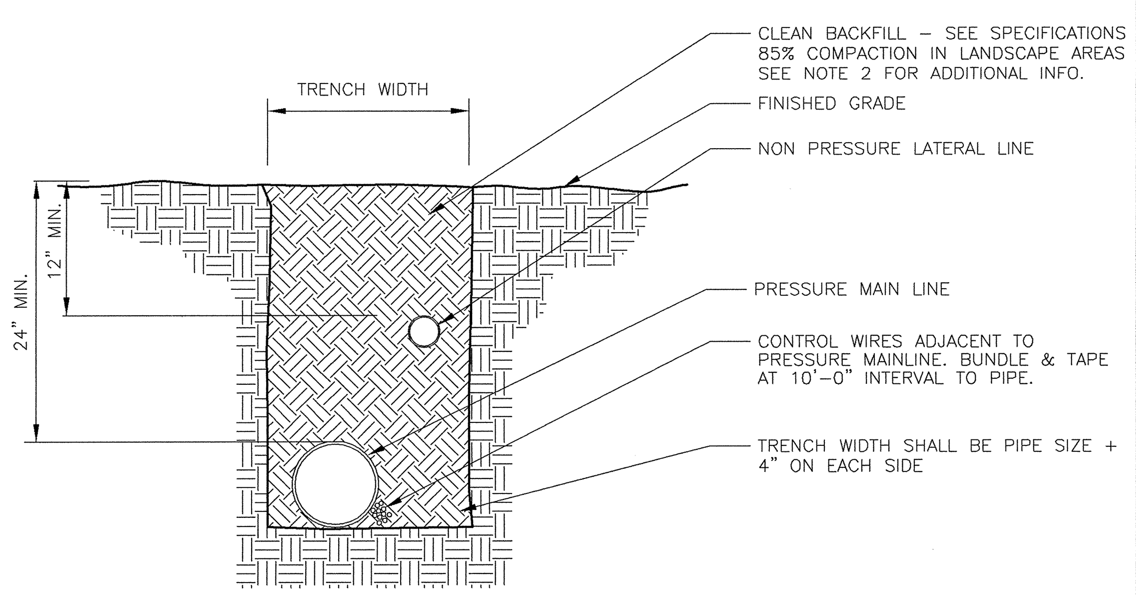
4 SUBSURFACE DRIP - ELBOW START CONNECTION DETAIL



5 SUBSURFACE DRIP - TEE START CONNECTION DETAIL

NOTES:

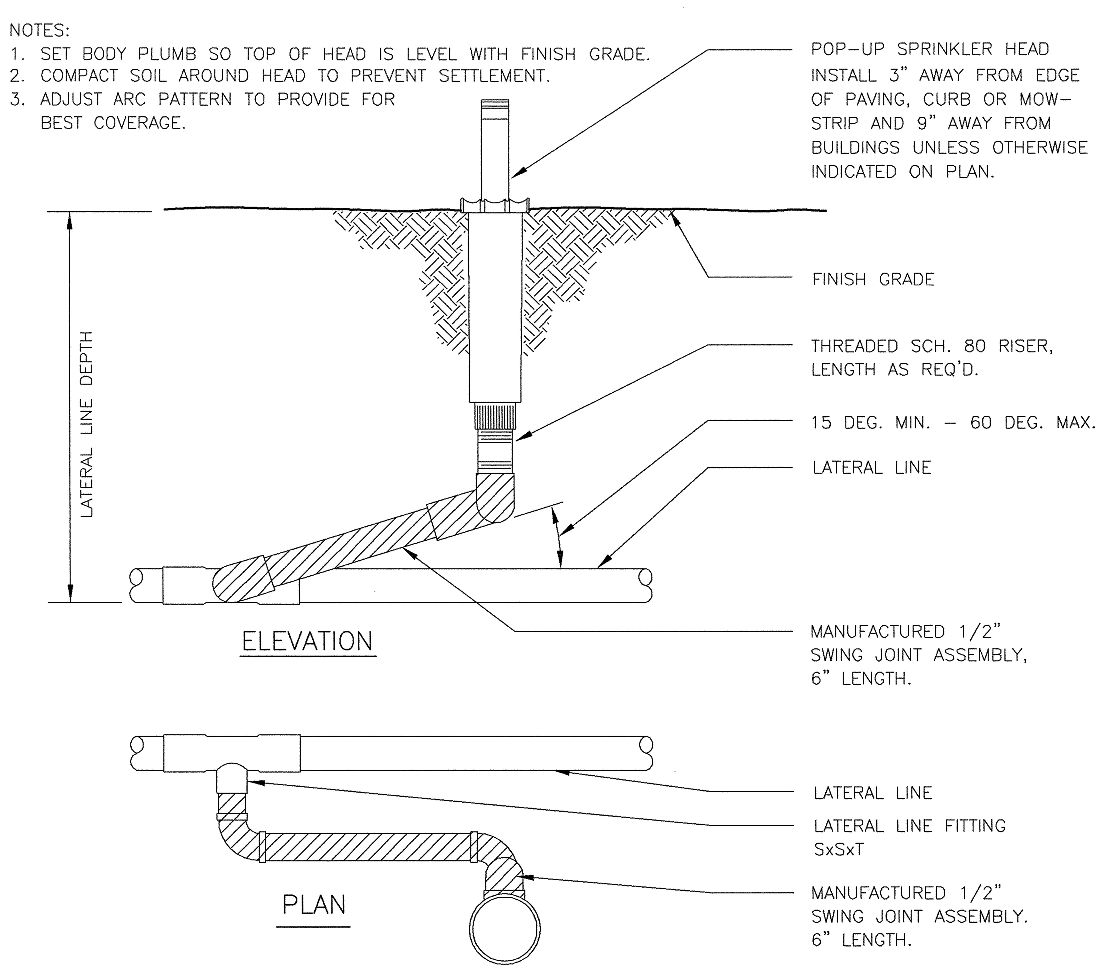
1. REFER TO SPECIFICATIONS AND PLAN SHEETS FOR MORE INFORMATION.
2. WHEN TRENCHES ARE LOCATED UNDER PAVEMENT, COMPACTION RATE FOR THE BACKFILL SHALL COMPLY WITH THE COMPACTION RATES REQUIRED FOR THOSE PAVING SECTIONS.



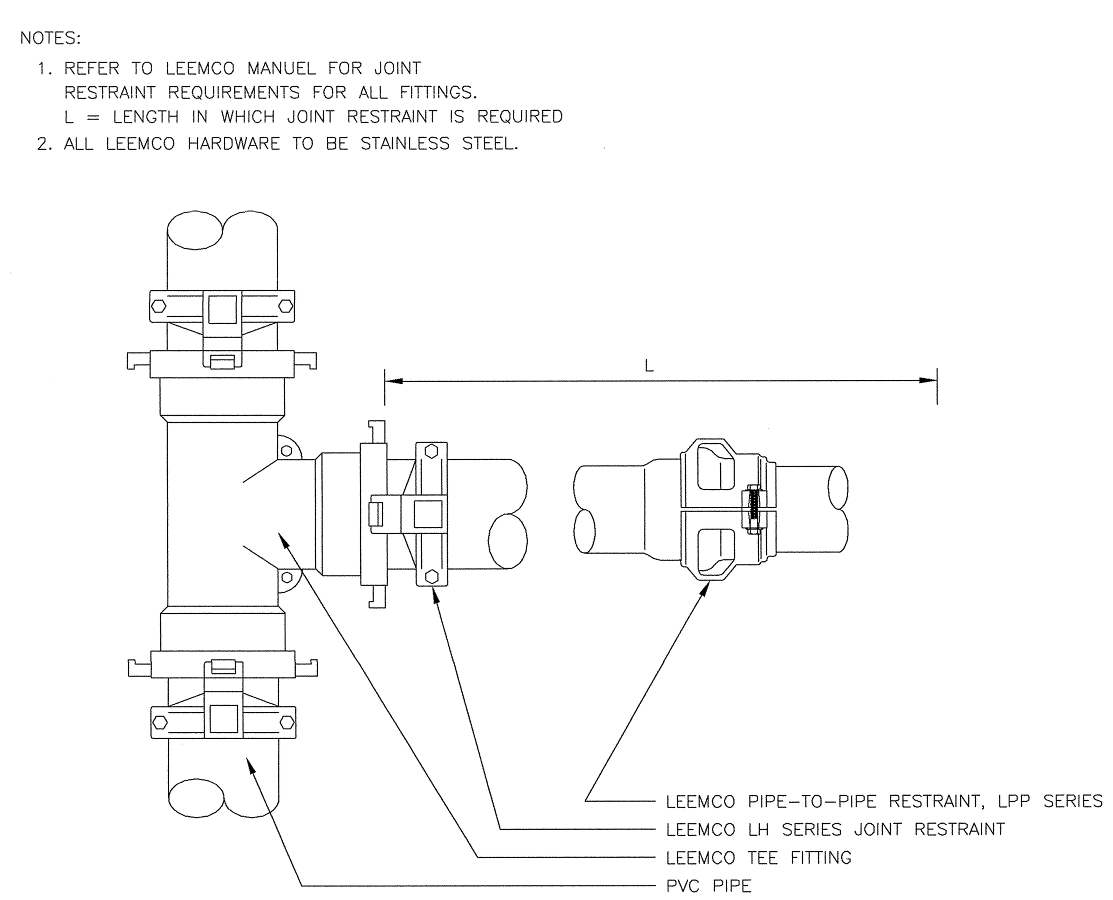
6 PIPE TRENCH DETAIL

INTENTIONALLY LEFT BLANK

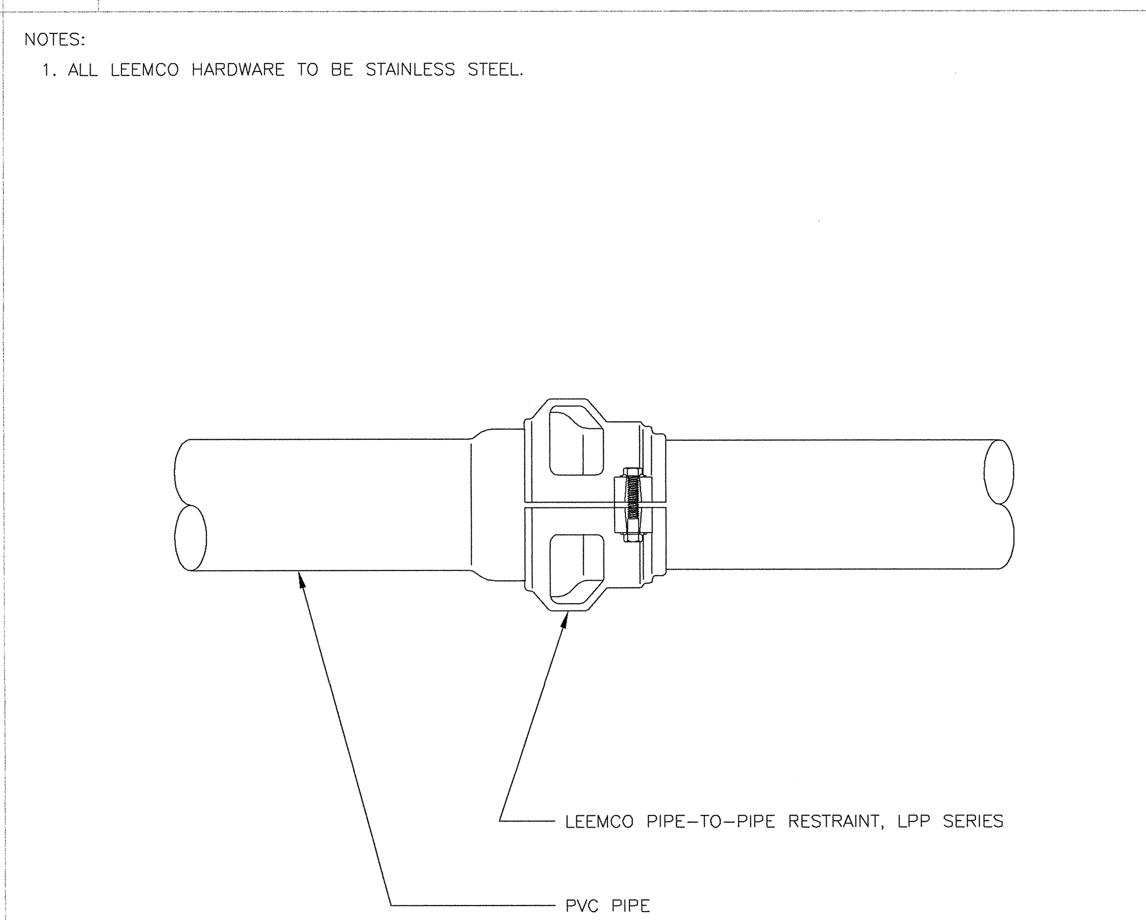
7



8 POP-UP SPRINKLER HEAD DETAIL



9 LEEMCO TEE WITH JOINT RESTRAINTS DETAIL



10 LEEMCO BELL JOINT RESTRAINTS DETAIL

DISTANCE CHART

REFER TO THE FOLLOWING TABLE THAT LISTS THE LENGTH (IN FEET) FOR EACH SIZE/TYPE FITTING WITHIN WHICH ALL JOINTS JUST BE RESTRAINED. ALL FITTINGS AND JOINT RESTRAINTS SHALL BE INSTALLED PER MANUFACTURERS RECOMMENDATIONS & SPECIFICATIONS.

AS AN EXAMPLE, IF YOU ARE INSTALLING A 3" MAINLINE WITH A DIRECTIONAL CHANGE OF 90°, REFER TO CHART UNDER PIPE SIZE TO 3" AND UNDER BENDS 90 YOU WILL SEE THE DISTANCE OF 11'. IF THERE IS ANY JOINT (VALVE, BELL, ETC.) YOU MUST INSTALL A JOINT RESTRAINT WITHIN 11' OF THE 90° MAINLINE DIRECTIONAL CHANGE.

PIPE SIZE	BENDS				REDUCERS			DEAD END	
	11°	22°	45°	90°	1 STEP	2 STEP	3 STEP	BLIND	SERV. B.
2"	1'	1'	2'	6'	–	–	–	19'	6'
2.5"	1'	2'	4'	9'	4'	–	–	23'	10'
3"	2'	3'	6'	11'	8'	10'	–	30'	15'
4"	2'	4'	9'	20'	14'	20'	31'	45'	25'
6"	3'	6'	13'	29'	30'	40'	53'	63'	40'
8"	4'	8'	15	38'	33'	55'	63'	75'	70'
10"	5'	9'	19'	45'	36'	56'	75'	96'	90'
12"	5'	10'	21'	53'	38'	60'	83'	112'	110'

INSTALLATION CHART

REFER TO THE FOLLOWING TABLE WHICH LISTS THE NUMBER OF BOLTS, SIZE, AND TORQUE FOR EACH BOLT IN REFERENCE TO THE SIZE OF PIPE WHICH IS BEING RESTRAINED.

AS AN EXAMPLE, IF YOU HAVE A 3" PIPE, YOU WILL NEED 2 BOLTS THAT ARE 3/8 X 2.5" AND TIGHTEN THEM WITH A TORQUE WRENCH TO 20 FT-LBS.

PIPE SIZE	NO. BOLTS	BOLT SIZE	TORQUE FT-LBS.
2"	2	3/8" x 2.5"	20
2.5"	2	3/8" x 2.5"	20
3"	2	3/8" x 2.5"	20
4"	2	1/2" x 3"	50
6"	2	1/2" x 3.5"	50
8"	4	1/2" x 4"	50
10"	4	5/8" x 5.5"	100
12"	4	5/8" x 5.5"	100

CONTACT NUNZIO DICHRISTOPHER @ (916) 202-1333, THE LEEMCO REPRESENTATIVE, FOR ALL QUESTIONS CONCERNING LEEMCO PRODUCTS. COORDINATE AN INSTALLATION CLINIC WITH NUNZIO PRIOR TO INSTALLING THE MAINLINE.

11 LEEMCO JOINT RESTRAINTS CHART

ONE INCH = TWENTY FEET
ONE-SIXTEENTH INCH = ONE FOOT
ONE-EIGHTH INCH = ONE FOOT
ONE-QUARTER INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE INCH = ONE FOOT
ONE AND ONE-HALF INCH = ONE FOOT

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JOB NO. Y1538.00

SHEET TITLE

Sprinkler Irrigation Details

SHEET NO.

L4.2

SHEET 6 OF 7 TOTAL

ONE AND ONE–HALF INCH = ONE FOOT
ONE INCH = ONE FOOT
THREE–QUARTERS INCH = ONE FOOT
ONE–HALF INCH = ONE FOOT
ONE–QUARTER INCH = ONE FOOT
ONE–EIGHTH INCH = ONE FOOT
ONE–SIXTEENTH INCH = ONE FOOT
ONE INCH = TWENTY FEET

LANDSCAPE HYDROZONE INFORMATION TABLE

STATION #/HYDROZONE	PLANT WATER USE TYPE	IRRIGATION TYPE	HYDROZONE AREA (HA) (SQ.FT.)	% OF TOTAL LANDSCAPE AREA
7	SHRUB - LOW	0.2	3,041	13.2%
8	SHRUB - LOW	0.2	636	2.8%
9	SHRUB - LOW	0.2	4,100	17.8%
10	SHRUB - LOW	0.2	1,489	6.4%
11	SHRUB - LOW	0.2	949	4.1%
12	SHRUB - LOW	0.2	7,781	33.7%
13	LAWN - HIGH (SLA)	0.8	5,091	22.1%
		TOTAL AREA	23,087	100.0%

IRRIGATION HYDROZONE INFORMATION TABLE

STATION #/HYDROZONE	PLANT WATER USE TYPE	PLANT FACTOR (PF)	HYDROZONE AREA (HA) (SQ.FT.)	PF x HA (SQ.FT.)	IRRIGATION EFFICIENCY (IE)	ETWU (GALLONS)
7	SHRUB - LOW	0.2	3,041	608.2	0.81	24,301
8	SHRUB - LOW	0.2	636	127.2	0.81	5,082
9	SHRUB - LOW	0.2	4,100	820.0	0.81	32,764
10	SHRUB - LOW	0.2	1,489	297.8	0.81	11,899
11	SHRUB - LOW	0.2	949	189.8	0.81	7,584
12	SHRUB - LOW	0.2	7,781	1556.2	0.81	62,179
13	LAWN - HIGH (SLA)	0.8	5,091	4072.8	0.72	183,072
		TOTAL AREA	17,996		ETWU TOTAL	326,880
		TOTAL AREA (SLA)	5091			
Eto (Roseville)	52.2					
ESTIMATED TOTAL WATER USAGE (ETWU) = (ETo)(0.62)(PF)(HA)/IE = GAL/YEAR						
MAXIMUM APPLIED WATER ALLOWANCE (MAWA) = (ETo)(0.62)[(0.7 x LA)+(0.3 x SLA)] = GAL/YEAR						
					MAWA TOTAL	442,379

IRRIGATION SCHEDULE TABLE

STATION #/HYDROZONE	PLANT WATER USE TYPE	PLANT FACTOR (PF)	IRRIGATION TYPE	FLOW (GPM)	PRECIP. RATE (PR) INCH/HR	IRRIGATION EFFICIENCY (IE)	SOIL TYPE	ROOT DEPTH	SLOPE	EXPOSURE
7	SHRUB - LOW	0.2	SUBSURFACE DRIP	13.5	0.43	0.81	SANDY LOAM	12-24"	0-5%	FULL SUN
8	SHRUB - LOW	0.2	SUBSURFACE DRIP	2.8	0.43	0.81	SANDY LOAM	12-24"	0-5%	FULL SUN
9	SHRUB - LOW	0.2	SUBSURFACE DRIP	18.2	0.43	0.81	SANDY LOAM	12-24"	0-5%	FULL SUN
10	SHRUB - LOW	0.2	SUBSURFACE DRIP	6.6	0.43	0.81	SANDY LOAM	12-24"	0-5%	FULL SUN
11	SHRUB - LOW	0.2	SUBSURFACE DRIP	4.2	0.43	0.81	SANDY LOAM	12-24"	0-5%	FULL SUN
12	SHRUB - LOW	0.2	SUBSURFACE DRIP	34.6	0.43	0.81	SANDY LOAM	12-24"	0-5%	FULL SUN
13	LAWN - HIGH	0.8	MP ROTATOR	33.3	0.45	0.72	SANDY LOAM	6"-9"	0-5%	FULL SUN

MAINTENANCE PERIOD (X/Y Z GAL)

STATION #/HYDROZONE	JANUARY	FEBUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMEBER	DECEMBER
7	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	25 /1 1,470 GAL	24 /2 2,919 GAL	21 /3 3,798 GAL	24 /3 4,312 GAL	21 /3 3,700 GAL	22 /2 2,647 GAL	20 /1 1,223 GAL	0 /5 0 GAL	0 /5 0 GAL
8	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	25 /1 305 GAL	24 /2 605 GAL	21 /3 788 GAL	24 /3 894 GAL	21 /3 768 GAL	22 /2 549 GAL	20 /1 254 GAL	0 /5 0 GAL	0 /5 0 GAL
9	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	25 /1 1,982 GAL	24 /2 3,936 GAL	21 /3 5,120 GAL	24 /3 5,813 GAL	21 /3 4,989 GAL	22 /2 3,568 GAL	20 /1 1,649 GAL	0 /5 0 GAL	0 /5 0 GAL
10	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	25 /1 719 GAL	24 /2 1,427 GAL	21 /3 1,857 GAL	24 /3 2,108 GAL	21 /3 1,809 GAL	22 /2 1,294 GAL	20 /1 598 GAL	0 /5 0 GAL	0 /5 0 GAL
11	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	25 /1 457 GAL	24 /2 908 GAL	21 /3 1,182 GAL	24 /3 1,342 GAL	21 /3 1,151 GAL	22 /2 823 GAL	20 /1 381 GAL	0 /5 0 GAL	0 /5 0 GAL
12	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	25 /1 3,767 GAL	24 /2 7,482 GAL	21 /3 9,734 GAL	24 /3 11,052 GAL	21 /3 9,484 GAL	22 /2 6,784 GAL	20 /1 3,135 GAL	0 /5 0 GAL	0 /5 0 GAL
13	0 /5 0 GAL	0 /5 0 GAL	0 /5 0 GAL	106 /1 15,591 GAL	105 /2 30,964 GAL	91 /3 40,285 GAL	104 /3 45,737 GAL	89 /3 39,250 GAL	95 /2 28,074 GAL	88 /1 12,974 GAL	0 /5 0 GAL	0 /5 0 GAL
MONTHLY RAINFALL (ROSEVILLE)	4.5	4.4	4.3	1.8	0.52	0.31	0.11	0.1	0.45	1.32	3.47	3.39
MONTHLY ET (ROSEVILLE)	1.1 JAN	1.7 FEB	3.1 MAR	4.7 APR	6.2 MAY	7.7 JUN	8.5 JUL	7.3 AUG	5.6 SEP	3.7 OCT	1.7 NOV	1.0 DEC
MONTHLY TOTALS (GAL)	0 GAL	0 GAL	0 GAL	24,290 GAL	48,241 GAL	62,765 GAL	71,258 GAL	61,151 GAL	43,740 GAL	20,214 GAL	0 GAL	0 GAL

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LICENSED LANDSCAPE ARCHITECT
BRYAN HOLLIS WALKER
5453
SIGNATURE
10/18/16
DATE
7/31/17
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CONSULTANT

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LICENSED ARCHITECT
BRIAN J. MAYTUM
No. C-26867
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SHEET TITLE

Sprinkler
Irrigation
Charts

SHEET NO.

L4.3

SHEET 7 OF 7 TOTAL

MEP COMPONENT ANCHORAGE NOTE

ALL MECHANICAL, PLUMBING, AND ELECTRICAL COMPONENTS SHALL BE ANCHORED AND INSTALLED PER THE DETAILS ON THE DSA APPROVED CONSTRUCTION DOCUMENTS. WHERE NO DETAIL IS INDICATED, THE FOLLOWING COMPONENTS SHALL BE ANCHORED OR BRACED TO MEET THE FORCE AND DISPLACEMENT REQUIREMENTS PRESCRIBED IN THE 2013 CBC, SECTIONS 1616A.1.18 THROUGH 1616A.1.26 AND ASCE 7–10 CHAPTER 13, 26 AND 30.

1. ALL PERMANENT EQUIPMENT AND COMPONENTS.
2. TEMPORARY OR MOVABLE EQUIPMENT THAT IS PERMANENTLY ATTACHED (e.g. HARD WIRED) TO THE BUILDING UTILITY SERVICES SUCH AS ELECTRICITY, GAS OR WATER.
3. MOVABLE EQUIPMENT WHICH IS STATIONED IN ONE PLACE FOR MORE THAN 8 HOURS AND HEAVIER THAN 400 POUNDS ARE REQUIRED TO BE ANCHORED WITH TEMPORARY ATTACHMENTS.

THE FOLLOWING MECHANICAL AND ELECTRICAL COMPONENTS SHALL BE POSITIVELY ATTACHED TO THE STRUCTURE, BUT THE ATTACHMENT NEED NOT BE DETAILED ON THE PLANS. THESE COMPONENTS SHALL HAVE FLEXIBLE CONNECTIONS PROVIDED BETWEEN THE COMPONENT AND ASSOCIATED DUCTWORK, PIPING, AND CONDUIT.

A. COMPONENTS WEIGHING LESS THAN 400 POUNDS AND HAVE A CENTER OF MASS LOCATED 4 FEET OR LESS ABOVE THE ADJACENT FLOOR OR ROOF LEVEL THAT DIRECTLY SUPPORT THE COMPONENT.
B. COMPONENTS WEIGHING LESS THAN 20 POUNDS, OR IN THE CASE OF DISTRIBUTION SYSTEMS, LESS THAN 5 POUNDS PER FOOT, WHICH ARE SUSPENDED FROM A ROOF OR FLOOR OR HUNG FROM A WALL.

FOR THOSE ELEMENTS THAT DO NOT REQUIRE DETAILS ON THE APPROVED DRAWINGS, THE INSTALLATION SHALL BE SUBJECT TO THE APPROVAL OF THE DESIGN PROFESSIONAL IN GENERAL RESPONSIBLE CHARGE OR STRUCTURAL ENGINEER DELEGATED RESPONSIBILITY AND THE DSA DISTRICT STRUCTURAL ENGINEER. THE PROJECT INSPECTOR WILL VERIFY THAT ALL COMPONENTS AND EQUIPMENT HAVE BEEN ANCHORED IN ACCORDANCE WITH ABOVE REQUIREMENTS.

PIPING, DUCTWORK & ELECTRICAL DISTRIBUTION SYSTEM BRACING NOTE

PIPING, DUCTWORK, AND ELECTRICAL DISTRIBUTION SYSTEMS SHALL BE BRACED TO COMPLY WITH THE FORCES AND DISPLACEMENTS PRESCRIBED IN ASCE 7–10 SECTION 13.3 AS DEFINED IN ASCE 7–10 SECTION 13.6.5.6, 13.6.7, 13.6.8, AND 2013 CBC, SECTIONS 1616A.1.23, 1616A.1.24, 1616A.1.25 AND 1616A.1.26.

THE METHOD OF SHOWING BRACING AND ATTACHMENTS TO THE STRUCTURE FOR THE IDENTIFIED DISTRIBUTION SYSTEM ARE AS NOTED BELOW. WHEN BRACING AND ATTACHMENTS ARE BASED ON PREAPPROVED INSTALLATION GUIDE (e.g., SMACNA OR OSHPD OPM). COPIES OF THE BRACING SYSTEM INSTALLATION GUIDE OR MANUAL SHALL BE AVAILABLE ON THE JOBSITE PRIOR TO THE START OF AND DURING THE HANGING AND BRACING OF THE DISTRIBUTION SYSTEMS. THE STRUCTURAL ENGINEER OF RECORD SHALL VERIFY THE ADEQUACY OF THE STRUCTURE TO SUPPORT THE HANGER AND BRACE LOADS.

MECHANICAL PIPING (MP), MECHANICAL DUCTS (MD), PLUMBING PIPING (PP), ELECTRICAL DISTRIBUTION SYSTEMS (E):

MP☐ MD☒ PP☐ E ☐ OPTION 1: DETAILED ON THE APPROVED DRAWINGS WITH PROJECT SPECIFIC NOTES AND DETAILS

MP☐ MD☒ PP☐ E ☐ OPTION 2: SHALL COMPLY WITH THE APPLICABLE OSHPD PRE–APPROVED (OPM #) #0043–13, 0052–13

MP☐ MD☒ PP☐ OPTION 3: SHALL COMPLY WITH THE SMACNA SEISMIC RESTRAINT MANUAL. OSHP EDITION (2009), INCLUDING ANY ADDENDA. FASTENERS AND OTHER ATTACHMENTS NOT SPECIFICALLY IDENTIFIED IN THE SMACNA SEISMIC RESTRAINT MANUAL, OSHPD EDITION, ARE DETAILED ON THE APPROVED DRAWINGS WITH PROJECT SPECIFIC NOTES AND DETAILS. THE DETAILS SHALL ACCOUNT FOR THE APPLICABLE SEISMIC HAZARD LEVEL "B" AND CONNECTION LEVEL "2" FOR THE PROJECT AND CONDITIONS.

MECHANICAL LEGEND

SYMBOL	ABBREVIATION	DESCRIPTION
	ABV	ABOVE
	ABC	ABOVE CEILING
	AF	ABOVE FLOOR
	AFF	ABOVE FINISHED FLOOR
	AFG	ABOVE FINISHED GRADE
	AD , AP	ACCESS DOOR , ACCESS PANEL
	AC	AIR CONDITIONING
	APD	AIR PRESSURE DROP, INCHES WATER COLUMN
	AB	ANCHOR BOLT
	ANV	ANGLE VALVE
	BV	BALL VALVE
	BDD	BACK DRAFT DAMPER
	BF	BELOW FLOOR
	BHP	BRAKE HORSE POWER
	BTU(H)	BRITISH THERMAL UNITS (PER HOUR)
BPT	BPT	BYPASS TIMER
	CBV	CALIBRATED BALANCE VALVE
	CC	CENTER TO CENTER
	CLG	CEILING
	CEF	CEILING EXHAUST FAN
	CKV	CHECK VALVE
	CLR	CLEAR
	CONC	CONCRETE
		CONCENTRIC REDUCER
—— CD ——	CD	CONDENSATE DRAIN
—— CR ——	CR	CONDENSATE RETURN
	COND	CONDENSER
—— CWS ——	CWS	CONDENSER WATER SUPPLY PIPING
– – – CWR – – –	CWR	CONDENSER WATER RETURN PIPING
	CONN	CONNECT OR CONNECTION
	CONT	CONTINUATION
	CONTR	CONTRACTOR
	CT	COOLING TOWER
f	CFM	CUBIC FEET OF AIR FLOW PER MINUTE
	DPR	DAMPER
°F		DEGREES FAHRENHEIT
ø	DIA	DIAMETER , PHASE
	DL	DOOR LOUVER
	DN	DOWN
		ECCENTRIC REDUCER
	EP	ELECTRICAL PANEL
	ENT	ENTERING
	EDB	ENTERING DRY BULB
	EW	ENTERING WATER
	EWT	ENTERING WATER TEMPERATURE
	EWB	ENTERING WET BULB
	EVAP	EVAPORATOR
	EC	EVAPORATIVE COOLER
	EA	EXHAUST AIR
	EAD	EXHAUST AIR DAMPER
	EF	EXHAUST FAN
	(E), EXIST	EXISTING
	ESP	EXTERNAL STATIC PRESSURE
	FPM	FEET PER MINUTE
	FIN	FINISH
	FC	FLEXIBLE CONNECTION
	FLR	FLOOR
		FLOW IN DIRECTION OF ARROW
	FLV	FLOW LIMITING VALVE
	FA	FROM ABOVE
	FB	FROM BELOW
	FLA	FULL LOAD AMPS
FS	FS	FIRE SMOKE DAMPER
	GALV	GALVANIZED
	GI	GALVANIZED IRON

MECHANICAL LEGEND cont'd

SYMBOL	ABBREVIATION	DESCRIPTION
	GA	GAUGE
	HTG	HEATING
	KW	KILOWATTS
	KWH	KILOWATT HOUR
	LDB	LEAVING DRY BULB IN DEGREES FAHRENHEIT
	LWB	LEAVING WET BULB IN DEGREES FAHRENHEIT
	LRA	LOCKED ROTOR AMPERES
	LVR	LOUVER
	MAD, MD	MANUAL AIR DAMPER
	MAV	MANUAL AIR VENT
	MFR	MANUFACTURER
	MAX	MAXIMUM
	MIN	MINIMUM
	MCC	MOTOR CONTROL CENTER
	OC	ON CENTER
	OA	OUTSIDE AIR
	OAD	OUTSIDE AIR DAMPER
	OD	OUTSIDE DIAMETER
	OV	OUTLET VELOCITY
	OH	OVERHEAD
	LBS	POUNDS
	PSI (G) (A)	POUNDS PER SQUARE INCH (GAUGE) (ABSOLUTE)
	PD	PRESSURE DROP
		PIPE DROP
		PIPE RISE
	PG	PRESSURE GAUGE
	PRV	PRESSURE REDUCING VALVE
—— RG ——	RG	REFRIGERANT GAS PIPING
—— RS ——	RS	REFRIGERANT SUCTION PIPING
—— RL ——	RL	REFRIGERANT LIQUID PIPING
	RA	RETURN AIR
	RAD	RETURN AIR DAMPER
	RPM	REVOLUTIONS PER MINUTE
	RLA	RUNNING LOAD AMPERES
	SB	SECURITY BARS
	SM	SHEET METAL
SD	SKD	SMOKE DETECTOR
	SQFT, FT^2	SQUARE FEET
	SQIN, IN^2	SQUARE INCHES
	SP	STATIC PRESSURE
	SPD	STATIC PRESSURE DROP
	STR	STRAINER
	SA	SUPPLY AIR
	SF	SUPPLY FAN
	TCP	TEMPERATURE CONTROL PANEL
	TCV	TEMPERATURE CONTROL VALVE
TS_X		TEMPERATURE SENSOR, "X" INDICATES DEVICE CONTROLLED
		THERMOMETER
T_X	T	THERMOSTAT, "X" TO TOP OF BOX INDICATES DEVICE CONTROLLED
	MBH	THOUSAND BRITISH THERMAL UNITS PER HOUR
	TA	TO ABOVE
	TB	TO BELOW
	TP	TOTAL PRESSURE
	TSP	TOTAL STATIC PRESSURE
	TYP	TYPICAL
	UG	UNDERGROUND
	UCD	UNDER CUT DOOR
	UON	UNLESS OTHERWISE NOTED
		UNION
	VLV	VALVE
		VALVE IN RISER (TYPE AS INDICATED OR NOTED)
		VALVE IN VALVE BOX
	WPD	WATER PRESSURE DROP
	W	WATTS
	WT	WEIGHT
	WB	WET BULB
	WMS	WIRE MESH SCREEN

DUCT LEGEND

SINGLE LINE SYMBOL	DOUBLE LINE SYMBOL	DESCRIPTION
24x12	24x12	RECTANGULAR DUCT – WIDTH x DEPTH (PLAN VIEW) DEPTH x WIDTH (SECTION VIEW)
26x14L	26x14L	ACOUSTICALLY LINED RECTANGULAR DUCT–DIMENSIONS ARE OUTSIDE
		MANUAL AIR DAMPER
R OR D	RISE OR DROP	RISE OR DROP DUCT IN DIRECTION OF AIR FLOW
	OR	RECTANGULAR TO RECTANGULAR TRANSISTION , MAX. SLOPE OF 1:3 RECTANGULAR TO ROUND TRANSITION , MAX. SLOPE OF 1:3
	W, R, $\frac{R}{W} = 1.5$	ELBOW, RECTANGULAR, SMOOTH RADIUS, WITHOUT TURNING VANES
		SQUARE/RECTANGULAR DUCT ELBOW WITH TURNING VANES
		CONVERGING OR DIVERGING TEE, 45° ENTRY, RECTANGULAR MAIN AND BRANCH. WHEN REDUCING MAIN, SIDE OF TAKE OFF OR ENTRY BRANCH TO BE FLAT, OTHER SIDES MAX. SLOPE OF 1:3
		CONICAL DUCT TAKE OFF FROM RECTANGULAR VIA SPIN–IN W/DAMPER AND SCOOP
	ROUND DUCT	ROUND DUCT TAKE OFF FROM RECTANGULAR VIA SMOOTH CONVERGING BELL MOUTH
		RECTANGULAR DUCT TEE MAD'S ON THE 2 BRANCHES, THROAT SIZED FOR EQUAL PRESSURE DROP
		RECTANGULAR DUCT SPLIT MAD'S, THROAT SIZED FOR EQUAL PRESSURE DROP
		3–WAY RECTANGULAR SPLIT WITH TWO TRANSITIONAL ELBOWS AND TRANSITIONING MAIN. DOWNSTREAM MAD'S ON THE TREE BRANCHES. THROATS SIZED FOR EQUAL PRESSURE DROP.
		FOR CONCEALED DUCT: DROP TO DIFFUSER SHALL BE FULL SIZE OF DIFFUSER NECK. FOR EXPOSED DUCT: DROP SHALL BE FULL SIZE OF OD DIFFUSER FRAME, FLANGE FOR MOUNITING DIFFUSER TURNED IN. AIR EXTRACTOR AND EQUALIZER GRID AT CONNECTION TO MAIN.
		SUPPLY AIR, SUPPLY AIR DUCT IN SECTION, SUPPLY DROP
		RETURN AIR, RETURN AND OUTSIDE AIR DUCT IN SECTION, RETURN AIR DROP
		EXHAUST AIR, EXHAUST AIR DUCT IN SECTION, EXHAUST AIR DROP
		FLEXIBLE DUCT (ROUND)
		FLEXIBLE DUCT (FABRIC)
MAD	18"ø, 10"ø, 12"ø	45° REDUCING LATERAL FITTING
18"ø, 10"ø, MAD	18"ø, 10"ø, 12"ø	90° REDUCING TEE FITTING

REFRIGERANT GENERAL NOTES

1. PER THE MANUFACTURER ALL REFRIGERANT PIPING RUNS BETWEEN HP–UNITS AND BC–EQUIPMENT SHALL BE RIGID, NO LINESETS.
2. PER THE MANUFACTURER ALL REFRIGERANT PIPING RUNS BETWEEN BC–EQUIPMENT AND FC–EQUIPMENT SHALL BE MANUFACTURED, INSULATED REFRIGERANT LINE–SET REFRIGERANT PIPING
3. PER THE MANUFACTURER ALL REFRIGERANT PIPING RUNS BETWEEN CU–EQUIPMENT AND F–EQUIPMENT SHALL BE MANUFACTURED, INSULATED REFRIGERANT LINE–SET REFRIGERANT PIPING

MECHANICAL GENERAL NOTES

1. ALL WORK SHALL COMPLY WITH ALL APPLICABLE CODES, SPECIFICATIONS, LOCAL ORDINANCES AND INDUSTRY STANDARDS.
2. VERIFY EXACT LOCATION OF ALL (E) EQUIPMENT, DUCTWORK, DIFFUSERS, REGISTERS AND GRILLES. NOTIFY ARCHITECT IMMEDIATELY OF ANY DISCREPANCIES BETWEEN (E) SYSTEMS AND DRAWINGS.
3. COORDINATE EXACT LOCATION OF EQUIPMENT AND ALL PENETRATIONS THROUGH ROOF, FLOORS AND WALLS WITH ARCHITECTURAL STRUCTURAL SYSTEMS PRIOR TO COMMENCING WORK.
4. COORDINATE EXACT SIZE AND ROUTING OF DUCTWORK WITH ARCHITECTURAL PLANS, STRUCTURE AND EQUIPMENT PRIOR TO COMMENCING WORK.
5. SEE ARCHITECTURAL REFLECTED CEILING PLAN FOR EXACT LOCATION OF ALL CEILING DIFFUSERS, REGISTERS AND GRILLES.
6. FURNISH AND INSTALL MANUAL AIR DAMPERS AT ALL DUCT BRANCH TAKEOFFS TO A SINGLE SUPPLY DUFFUSER.
7. FLEXIBLE DUCTWORK CONNECTIONS TO CEILING DIFFUSERS ARE LIMITED TO 8' MAXIMUM LENGTH.
8. ALL DUCTWORK, CEILING DIFFUSERS/REGISTERS/GRILLES, EQUIPMENT, PIPING ETC., ARE NEW U.O.N. (SHOWN HEAVY). (E) DUCTWORK, PIPING ETC. IS SHOWN LIGHT. SEE LEGEND.
9. (E) DUCTWORK AND ITEMS TO BE REMOVED ARE SHOWN CROSSED ("X") OUT, SEE LEGEND, COORDINATE CLOSELY WITH (N) DUCTWORK AND P.O.C.'S SHOWN. ALL OTHER (E) DUCTWORK, ETC. TO REMAIN.
10. WHERE INLET DUCT DIAMETER AND DIFFUSER NECK SIZE ARE THE SAME (I.E. 9"ø & 9x9) CONTRACTOR SHALL OVERSIZE THE SHEET METAL PLENUM TO ACCOMODATE THE ROUND DUCT CONNECTION.
11. THERMOSTATS AND ROOM TEMPERATURE SENSORS SHALL BE INSTALLED AT 46" ABOVE FINISHED FLOOR (TO TOP OF DEVICE). DO NOT INSTALL THERMOSTATS AND ROOM TEMPERATURE SENSORS ABOVE CASEWORK, SHELVING OR OTHER OBSTRUCTIONS OVER 24" IN DEPTH AND 34" IN HEIGHT.

Dry Creek Joint Elementary School District
District Education Center
8849 Cook Riolo Road, Roseville, CA 95747
Dry Creek Joint Elementary School District

FILE NO. 31-15

IDENTIFICATION STAMP DIV. OF THE STATE ARCHITECT OFFICE OF REGULATION SERVICES
APPLICATION NO. 02-115496
AC TM FLS HMR SS KR
DATE 2/23/17 CCA2

DIVISION OF THE STATE ARCHITECT
revision date: ________ by: ________

REGISTERED PROFESSIONAL ENGINEER THOMAS A. DUMAL M 22836 EXPIRES 9/30/18 MECHANICAL STATE OF CALIFORNIA
DATE SIGNED: 2/15/2017

CAPITAL ENGINEERING CONSULTANTS, INC.
RANCHO CORDOVA, CALIFORNIA
RL — PM - DESIGN TEAM
160110.00 — PROJECT NO.

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LICENSED ARCHITECT BRIAN J. MAY No. C-26867 RENEWAL DATE 1/31/19 STATE OF CALIFORNIA

ARCHITECT

INITIAL BOX

NO.	DWG BY	DATE	REVIEWED

REVISIONS

NO.	DESCRIPTION	DATE	REV'D
	100% design developement	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017
JOB NO. Y1538.00
SHEET TITLE

Mechanical Schedules, Legends and Notes

SHEET NO.

M0.1

SHEET____OF____TOTAL

MISC. EQUIPMENT SCHEDULE

UNIT	DESCRIPTION
CF 1, CF 2, CF 3	**CEILING FAN - HIGH VELOCITY LOW SPEED FAN** "MACROAIR" MODEL AIRVOLUTION–D 370 CEILING FAN. 38,000 CFM. 8 FOOT TOTAL BLADE DIAMETER. MOUNT AT 14'–0" ABOVE FINISHED FLOOR. PROVIDE WITH MOUNTING EXTENSION TO MEET A 14'–0" AFF MOUNTING. PROVIDE ALL MOUNTING HARDWARE AND CONTROLS. FAN SHALL BE INTERLOCKED WITH FIRE PROTECTION AND SHUT OFF UPON ALARM. AIRVOLUTION–D 370 DIGITAL TOUCH PAD CONTROLLER. BOTTOM OF BLADE SHALL BE AT 14'–0" ABOVE FINISHED FLOOR. ELECTRICAL: 120V/1 PH., 0.25 HP, 270 WATTS, 5.9 AMPS. RECOMMENDED FUSE SIZE = 10 AMPS. OPER. WEIGHT = 59 LBS. FOR MOUNTING SEE DETAIL: 1/M5.1 FOR CONTROL DIAGRAM SEE DETAIL: 1/M7.1

DIFFUSER, REGISTER & GRILLE SCHEDULE

SYMBOL	DESCRIPTION	KRUEGER	METALAIRE	NAILOR	TITUS	TUTTLE & BAILEY
CD	MODULAR CORE SURFACE MOUNT CEILING DIFFUSER BEVEL FRAME $\frac{3}{4}$" DROP	1240 FRAME 21 – $1\frac{1}{4}$"	9000–2	7500–S	MCD BORDER TYPE 6	SQD–SB
CD–2	MODULAR CORE SURFACE MOUNT CEILING DIFFUSER FLAT FRAME	1240 FRAME 22	9000–1	7500–B	MCD BORDER TYPE 1	SQD–SF
CDL	MODULAR CORE LAY–IN CEILING DIFFUSER FOR T–BAR CEILING 24x24 PANEL	1240 FRAME 23	9000–6P	7500–L	MCD BORDER TYPE 3	SQD–LT
CR	CEILING RETURN WITH $\frac{1}{2}$" EGG CRATE CORE SURFACE MOUNT	EGC–5	CC5D	61 EC–S	MODEL 50 F BORDER TYPE 1	CRE500–SF
CRL	CEILING RETURN WITH $\frac{1}{2}$" EGG CRATE CORE IN 24x24 PANEL FOR T–BAR CEILING	EGC–5TB	CC5D–TBD	61 EC–L	MODEL 50 F BORDER TYPE 3	CRE500–LT
S **	DOUBLE DEFLECTION SUPPLY GRILLE WITH VERTICAL FRONT BARS, $\frac{3}{4}$" SPACING. PROVIDE WITH MILL FINISH 04 ALUMINUM FINISH.	880 V	V 4004 S	61 DV	300 RS	T54
R & E *	RETURN OR EXHAUST GRILLE WITH 35° OR 45° HORIZONTAL BARS.	S 80 H	SRH	7145 H	350 RL	T70D
SG	SOFFIT GRILLE – HEAVY DUTY SINGLE DEFLECTION GRILLE WITH 10 GAUGE, $\frac{3}{8}$" WOVEN STEEL MESH SECURED BEHIND FACE BARS. PROVIDE PLASTER FRAME IN PLASTER SOFFIT	S 480 H WITH $\frac{3}{8}$" MESH AND PF WHERE REQUIRED	HDRH WITH $\frac{3}{8}$" MESH AND PF WHERE REQUIRED	6145 HD WITH $\frac{3}{8}$" MESH & PLASTER FRAME WHERE REQUIRED	33 RL HD WITH $\frac{3}{8}$" MESH AND PF WHERE REQUIRED	T75D WITH $\frac{3}{8}$" MESH AND PF WHERE REQUIRED
RH & EH	HEAVY DUTY RETURN OR EXHAUST GRILLE WITH 35° OR 45° HORIZONTAL BARS	S 480 H	HDRH	6145 HD	33 RL	T115H–40
LD	ALUMINUM LINEAR SLOT DIFFUSER WITH 4–$\frac{3}{4}$" SLOTS & FIELD FABRICATED PLENUM	–	–	–	ML–38	6000
LD–2	ALUMINUM LINEAR SLOT DIFFUSER WITH 8–$\frac{3}{4}$" SLOTS & FIELD FABRICATED PLENUM	–	–	–	ML–38	6000
TFL	"ACCUTHERM" THERMA–FUSER ST–HC THERMALLY POWERED VAV DIFFUSER, FOR 24x24 LAY–IN T–BAR CEILING	N/A	N/A	N/A	N/A	N/A

NOTES:
1. ALL SYMBOLS NOTED MAY NOT BE USED. REFER TO PLANS FOR SIZE AND QUANTITY.
2. ALL SUPPLY AIR DIFFUSERS ARE 4 WAY BLOW UNLESS SHOWN OTHERWISE.
3. FURNISH ALL PRODUCTS OF A SINGLE MANUFACTURER.
4. COORDINATE DIFFUSER TYPE WITH REFLECTED CEILING PLAN.
5. OPPOSED BLADE DAMPERS ARE NOT REQUIRED AT DIFFUSERS, REGISTERS OR GRILLES.
6. PROVIDE MANUAL AIR DAMPERS AT EACH BRANCH DUCT TO A SINGLE DIFFUSER, REGISTER OR GRILLE.

* ALUMINUM REGISTERS FOR SHOWERS AND DAMP AREAS

** ALUMINUM WITH MILL FINISH 04 ALUMINUM FINISH

FURNACE SCHEDULE (WITH DX COOLING COIL)

UNIT	SERVES	"CARRIER" MODEL NO. U.N.O.	CFM	MIN. OSA (CFM)	ESP (IN. WG.)	DX COOLING COIL				GAS HEATING			ELECTRICAL DATA						AFUE (%)	OPER. WT. (LBS.)	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
						"CARRIER" MODEL NO.	APD, WET (IN. WG.)	EDB (°F)	EWB (°F)	INPUT (MBH)	OUTPUT (MBH)	EDB (°F)	VOLT/PH	SUPPLY FAN HP	SUPPLY FAN RPM	SUPPLY FAN FLA	MCA	MOCP					
F 1	COMMUNITY ROOM–A 103	59SP5A060–14	1350	210	0.5	CNPVP3617ALA	0.38	80.0	62.0	60.0	58.0	62.53	115/1ø	3/4	1505 MAX.	9.9	12.5	15	96.5	48.5 141 59	2/M5.3	6/M7.1	1 2 3 4 5 6 7
F 2	COMMUNITY ROOM–B 104	59SP5A080–20	1800	275	0.5	CNPVP4821ALA	0.318	80.0	62.0	80.0	78.0	62.67	115/1ø	1.0	1900 MAX.	12.3	15	20	96.5	65.5 155.5 81	2/M5.3	6/M7.1	1 2 3 4 5 6 7
F 3	PUBLIC LOBBY 100	59SP5A080–20	1800	430	0.5	CNPVP4821ALA	0.318	80.0	62.0	80.0	78.0	58.53	115/1ø	1.0	1900 MAX.	12.3	15	20	96.5	65.5 155.5 81	5/M5.3	6/M7.1	1 2 3 4 5 6 7
F 4	OPEN OFFICE 110	59SP5A080–20	1800	230	0.5	CNPVP4821ALA	0.318	80.0	62.0	80.0	78.0	63.8	115/1ø	1.0	1900 MAX.	12.3	15	20	96.5	65.5 155.5 81	2/M5.3	6/M7.1	1 2 3 4 5 6 7
F 5	PRE–FUNCTION HALLWAY 102	59SP5A040–10	700	305	0.5	CNPVP2414ALA	0.21	80.0	62.0	40.0	39.0	49.09	115/1ø	1/2	925 MAX.	6.8	8	15	96.5	39.5 120 46	2/M5.3	6/M7.1	1 2 3 4 5 6 7
F 6	OPEN OFFICE SOUTH 110	59SP5A080–20	1800	230	0.5	CNPVP4821ALA	0.318	80.0	62.0	80.0	78.0	63.87	115/1ø	1.0	1900 MAX.	12.3	15	20	96.5	65.5 155.5 81	2/M5.3	6/M7.1	1 2 3 4 5 6 7
F 7	OPEN OFFICE NORTH 126	59SP5A080–20	1575	195	0.5	CNPVP4221ALA	0.35	80.0	62.0	80.0	78.0	64.06	115/1ø	1.0	1900 MAX.	12.3	15	20	96.5	56.5 155.5 81	2/M5.3	6/M7.1	1 2 3 4 5 6 7
F 8	OPEN OFFICE NORTH 126	59SP5A120–22	1909	245	0.5	CNPVP6024ALA	0.23	80.0	62.0	120.0	117.0	63.84	115/1ø	1.0	1900 MAX.	11.1	15	20	96.5	78.0 189.5 81	2/M5.3	6/M7.1	1 2 3 4 5 6 7

NOTES:
1. UNIT SELECTED AT 102°F DB / 70°F WB SUMMER AMBIENT, AND 22°F DB WINTER AMBIENT AIR TEMPERATURES.
2. ESP SCHEDULED DOES NOT INCLUDE DX COIL APD.
3. SEE CONDENSING UNIT SCHEDULE FOR COOLING CAPACITIES.
4. OPERATING WEIGHT INCLUDES DX COIL.
5. PROVIDE WITH MANUFACTURER'S CONCENTRIC COMBUSTION AIR/VENT KIT. SEE DETAIL 1/M5.3.
6. PROVIDE WITH 1" MERV 8 DISPOSABLE PLEATED MEDIA FILTERS.
7. LOWER OUTSIDE AIR POSITION INDICATED IS BASED ON 0.15 CFM/SQ.FT., ALLOWABLE FOR CO2 DEMAND CONTROLLED SYSTEMS AT MINIMUM OCCUPANCY. UPPER OUTSIDE AIR POSITION INDICATED IS BASED ON 15 CFM/OCCUPANT WHEN SPACE IS AT MAXIMUM OCCUPANCY, UNLESS SYSTEM IS IN ECONOMIZER MODE. SEE CONTROLS FOR SEQUENCE OF OPERATION. FOR THESE UNITS WITH CO2 CONTROL, ENTERING TEMPERATURES SCHEDULED REPRESENT CONDITIONS AT UPPER OSA POSITION.

CONDENSING UNIT SCHEDULE

UNIT	SERVES	"CARRIER" MODEL NO.	EVAP. CFM	SENSIBLE COOLING CAP. (MBH)	TOTAL COOLING CAP. (MBH)	REFRIG. PIPE		ELECTRICAL DATA						SEER (EER)	OPER. WT. (LBS.)	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
						RS (DIA.)	RL (DIA.)	VOLT/PH	COND. FAN FLA	COMPRESSOR RLA	COMPRESSOR LRA	MCA	MOCP					
CU 1	COMMUNITY ROOM–A 103	24AAA536A003	1350	30.98	30.98	3/4	3/8	208/1ø	1.10	13.6	79.0	18.1	30	14.5	205	10/M5.2	6/M7.1	1 2
CU 2	COMMUNITY ROOM–B 104	24AAA548A003	1800	39.36	39.36	7/8	3/8	208/1ø	1.40	15.6	105.7	20.9	35	14.5	240	10/M5.2	6/M7.1	1 2
CU 3	PUBLIC LOBBY 100	24AAA548A003	1800	39.36	39.36	7/8	3/8	208/1ø	1.40	15.6	105.7	20.9	35	14.5	240	10/M5.2	6/M7.1	1 2
CU 4	OPEN OFFICE 110	24AAA548A003	1800	39.36	39.36	7/8	3/8	208/1ø	1.40	15.6	105.7	20.9	35	14.5	240	10/M5.2	6/M7.1	1 2
CU 5	PRE–FUNCTION HALLWAY 102	24AAA524A003	700	20.42	20.42	5/8	5/16	208/1ø	0.50	10.9	62.9	14.1	20	14.5	175	10/M5.2	6/M7.1	1 2
CU 6	OPEN OFFICE SOUTH 110	24AAA548A003	1800	39.36	39.36	7/8	3/8	208/1ø	1.40	15.6	105.7	20.9	35	14.5	240	10/M5.2	6/M7.1	1 2
CU 7	OPEN OFFICE NORTH 126	24AAA542A003	1575	36.98	36.98	3/4	3/8	208/1ø	1.40	16.7	109.0	22.3	35	14.5	270	10/M5.2	6/M7.1	1 2
CU 8	OPEN OFFICE NORTH 126	24AAA560A003	1909	49.27	49.27	7/8	3/8	208/1ø	4.52	20.8	124.1	27.5	40	17.0	260	10/M5.2	6/M7.1	1 2

NOTES:
1. R–410A REFRIGERANT.
2. PROVIDE WITH EXPANDED METAL CONDENSER COIL GUARDS.

FAN SCHEDULE

UNIT	LOCATION	"MFR" MODEL NO.	CFM	ESP (IN. W.G.)	DUTY	STYLE	FLA	RPM	HP	VOLT/PH	OPER. WT. (LBS.)	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
IEF 1	ATTIC	GREENHECK MODEL SQ–99–VG	710	0.6	E	IL	11.3	1778	3/4	115/60/1	70	6&7/M5.1	2/M7.1	INTERLOCK WITH F–3 1
IEF 2	ATTIC	GREENHECK MODEL SQ–99–VG	710	0.6	E	IL	11.3	1778	3/4	115/60/1	70	6&7/M5.1	2/M7.1	INTERLOCK WITH F–5 1
IEF 3	ATTIC	GREENHECK MODEL SQ–97–VG	175	0.5	E	IL	3.7	1479	1/4	115/60/1	60	6&7/M5.1	3/M7.1	INTERLOCK WITH LIGHTING 1

LEGEND
DUTY: S–SUPPLY, R–RETURN, E–EXHAUST
STYLE: BI–BACKWARD INCLINED, FC–FORWARD CURVED, AF–AIRFOIL, RD–RADIAL, TU–TUBULAR, IL–INLINE, VS–VENT SET, VA–VANE AXIAL, TA–TUBE AXIAL, PP–PROPELLAR, RE–ROOF EXHAUST, WE–WALL EXHAUST, CA–CABINET, CE–CEILING

NOTES:
1. PROVIDE WITH THERMAL OVERLOAD PROTECTION.

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AC TM FLS HMR SS KR
DATE 2/23/17 CCA2

DIVISION OF THE STATE ARCHITECT
revision date: ________ by: ________

REGISTERED PROFESSIONAL ENGINEER THOMAS A. DUNN M 22836 EXPIRES 9/30/18 MECHANICAL STATE OF CALIFORNIA
DATE SIGNED: 2/15/2017

CAPITAL ENGINEERING CONSULTANTS, INC.
RANCHO CORDOVA, CALIFORNIA
RL — PM - DESIGN TEAM
160110.00 — PROJECT NO.

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LICENSED ARCHITECT BRIAN J. MAYTUM No. C-26867 RENEWAL DATE: 1/31/19 STATE OF CALIFORNIA

ARCHITECT

INITIAL BOX

NO.	DWG BY	DATE	REVIEWED

REVISIONS

NO.	DESCRIPTION	DATE	REVD
	100% design developement	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017
JOB NO. Y1538.00
SHEET TITLE
Mechanical Schedules
SHEET NO.
M0.2
SHEET ____ OF ____ TOTAL

FAN COIL UNIT SCHEDULE

UNIT	SPACE SERVED	"MITSUBISHI" MODEL NO. (INDOOR UNIT)	CFM	MIN. OA (CFM)	ESP (IN. WG)	SENSIBLE COOLING CAPACITY (BTUH)	NOMINAL COOLING CAPACITY (BTUH)	NOMINAL HEATING CAPACITY (BTUH)	ELECTRICAL DATA VOLT/PH	FAN MOTOR	AMPS	MCA	MOCP (AMPS)	CONNECTED TO OUTDOOR UNIT	CONNECTED TO BC CONTROLLER	OPER. WT. (LBS.)	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
FC 1	ASSIST. SUP. ADMIN. RESOURCES 133	PMFY-P06NBMU-E	230 TO 307	27	0.0	4,500	6,000	6,700	208V/1PH	1	0.20	0.25	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 2	HUMAN RESOURCES DIRECTOR 132	PMFY-P06NBMU-E	230 TO 307	26	0.0	4,500	6,000	6,700	208V/1PH	1	0.20	0.25	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 3	CONFERENCE ROOM 131	PMFY-P12NBMU-E	258 TO 328	85	0.0	XX	12,000	13,500	208V/1PH	1	0.21	0.26	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 4	CBO OFFICE 125	PMFY-P06NBMU-E	230 TO 307	28	0.0	4,500	6,000	6,700	208V/1PH	1	0.20	0.25	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 5	OFFICE 124	PMFY-P12NBMU-E	258 TO 328	86	0.0	8,175	9,450	13,500	208V/1PH	1	0.21	0.26	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 6	OFFICE 123	PMFY-P12NBMU-E	258 TO 328	86	0.0	8,175	9,450	13,500	208V/1PH	1	0.21	0.26	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 7	CONFERENCE ROOM 122	PMFY-P12NBMU-E	258 TO 328	86	0.0	8,175	9,450	13,500	208V/1PH	1	0.21	0.26	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 8	IDF 128	PMFY-P06NBMU-E	230 TO 307	17	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 9	ED. SERVICES CONF. RM. 102	PLFY-P30NBMU-E	565 TO 777	198	0.0	20,550	23,625	34,000	208V/1PH	1	0.51 HTG 0.43 CLG	0.64	15A	HP 2	BC 2	64	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 10	ES-1 OFFICE 120	PMFY-P06NBMU-E	230 TO 307	23	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 11	ES-2 OFFICE 119	PMFY-P06NBMU-E	230 TO 307	22	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 12	ES-3 OFFICE 118	PMFY-P06NBMU-E	230 TO 307	22	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 13	ES-4 OFFICE 117	PMFY-P06NBMU-E	230 TO 307	22	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 14	ES-6 OFFICE 115	PMFY-P06NBMU-E	230 TO 307	23	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 15	ASSIST. SUP. ED. SERVICES 114	PMFY-P06NBMU-E	230 TO 307	31	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 16	OFFICE 113	PMFY-P06NBMU-E	230 TO 307	27	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 1	BC 1	38	9 M5.1	1 M7. SERIES	1 2 3 6 8
FC 17	SUPERINT. CONF. RM. 112	PLFY-P30NBMU-E	565 TO 777	171	0.0	20,550	23,590	34,000	208V/1PH	1	0.51 HTG 0.43 CLG	0.64	15A	HP 1	BC 1	64	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 18	SUPERINT. OFFICE 111	PLFY-P12NCMU-E	320 TO 390	43	0.0	9,150	9,169	13,500	208V/1PH	1	0.28	0.35	15A	HP 1	BC 1	44	9 M5.1	1 M7. SERIES	1 3 4 5 6 7 8
FC 19	ED SERVICES WORKROOM 127	PLFY-P08NCMU-E	280 TO 350	53	0.0	6,300	9,169	9,000	208V/1PH	1	0.23	0.29	15A	HP 2	BC 2	41	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 20	CONFERENCE ROOM 131	PMFY-P15NBMU-E	272 TO 378	85	0.0	9,150	11,800	17,000	208V/1PH	1	0.26	0.33	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 7 8
FC 21	ES-5 OFFICE 116	PMFY-P06NBMU-E	230 TO 307	22	0.0	4,725	4,725	6,700	208V/1PH	1	0.20	0.25	15A	HP 2	BC 2	38	9 M5.1	1 M7. SERIES	1 2 3 6 8

NOTES: MITSUBISHI VRF SYSTEM AND CONTROLS TO MATCH DRY CREEK UNIFIED SCHOOL DISTRICT CORPORATION YARD STANDARD. NO OTHER MANUFACTURERS TO BE CONSIDERED.

1. FAN COIL OPERATING WT. INCLUDES FAN COIL AND INCLUDED COMPONENTS MOUNTED.
2. CONNECT OUTSIDE AIR SUPPLY SYSTEM AS SHOWN ON PLANS
3. PROVIDE WITH WALL MOUNTED CONTROLLED THERMOSTAT.
4. PROVIDE FACTORY MIXING BOX w/ ACTUATED RETURN AND OUTSIDE AIR DAMPERS.
5. PROVIDE 2" PANEL FILTER, MIN. SIZE OF 18"x20"x2" (TRI-PLEX 85 OR EQUAL). FILTER TO BE INSTALLED IN FLAT RACK AND ACCESS DOOR FOR REMOVAL. FILTERS SHALL BE "MERV 8 OR GREATER".
6. REFER TO PIPING AND IDENTIFICATION DWGS. M7.2, M7.3, M7.4, M7.4 FOR REFRIGERANT PIPING SIZES BETWEEN FAN COIL AND BRANCH CONTROLLER.
7. PROVIDE WITH PAC-SHT53TM-E MULTI-FUNCTIONAL CASEMENT AND PAC -SH59KF-E MERV 10 FILTER ACCESSORIES FOR OUTSIDE AIR INTAKE.
8. MODIFY T-BAR CEILING TO ACCOMODATE FC COMPLETE WITH TRIM PANEL.

GENERAL NOTES

1. PER THE MANUFACTURER ALL REFRIGERANT PIPING RUNS BETWEEN HP-UNITS AND BC-EQUIPMENT SHALL BE RIGID, NO LINESETS.
2. PER THE MANUFACTURER ALL REFRIGERANT PIPING RUNS BETWEEN BC-EQUIPMENT AND FC-EQUIPMENT SHALL BE MANUFACTURED, PRE-CHARGED AND PRE-INSULATED REFRIGERANT LINE-SET REFRIGERANT PIPING
3. PER THE MANUFACTURER ALL REFRIGERANT PIPING RUNS BETWEEN CU-EQUIPMENT AND F-EQUIPMENT SHALL BE MANUFACTURED, PRE-CHARGED AND PRE-INSULATED REFRIGERANT LINE-SET REFRIGERANT PIPING

VRF HEAT PUMP UNIT SCHEDULE

UNIT	SERVES	"MITSUBISHI" MODEL NO. (OUTDOOR UNIT)	NOMINAL COOLING CAPACITY (MBH)	NOMINAL HEATING CAPACITY (MBH)	REFRIG. PIPE RS (DIA.)	REFRIG. PIPE RL (DIA.)	ELECTRICAL DATA VOLT/PH	COOL KW	HEAT KW	MCA	MAX. FUSE	IEER	OPER. WT. (LBS.)	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
HP 1	BLDG A ADMIN	PURY-P72TLMU-A	72.0	80.0	3/4"	5/8"	208V/3 PH	21.1	23.4	24	35	23.1/ 28.1	444	9 M5.2	1 M7.2, 1 M7.4	1 2 3 4
HP 2	BLDG A OFFICES	PURY-P120TLMU-A	120.0	135.0	1-1/8"	3/4"	208V/3 PH	35.2	39.6	45	60	18.6 20.8	715	9 M5.2	1 M7.3, 1 M7.5	1 2 3 4

NOTES: MITSUBISHI VRF SYSTEM AND CONTROLS TO MATCH DRY CREEK UNIFIED SCHOOL DISTRICT CORPORATION YARD STANDARD. NO OTHER MANUFACTURERS TO BE CONSIDERED.

1. UNITS SELECTED AT 105 DEG. F DB / 70 DEG. F WB SUMMER AMBIENT, 30 DEG. F DB WINTER AMBIENT AIR TEMPERATURES.
2. R-410A REFRIGERANT.
3. SIMULTANEOUS HEATING AND COOLING OPERATION.
4. PROVIDE SYSTEM WITH FACTORY BACNET INTERFACE MODULE.

VRF BRANCH CIRCUIT CONTROLLER SCHEDULE

UNIT	SERVES	ASSOC. OUTDOOR UNIT	DESCRIPTION	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
BC 1	DEC BLDG	HP/1	"MITSUBISHI" MODEL #CMB-P1010NU-GA MAIN BRANCH CIRCUIT CONTROLLER, 10 TOTAL REFRIGERANT BRANCH CIRCUITS, 208V/1ph, COOLING: 0.183/0.236 KW POWER INPUT, 0.88/ 1.03 CURRENT INPUT, HEATING: 0.092/0.118 POWER INPUT KW, 0.45/ 0.52 CURRENT INPUT AMPS, 1.60/1.88 MCA, 15 AMPS MOCP. OPERATING WEIGHT: 160 LBS.	8 M5.1	1 M7.2, 1 M7.4	1 2
BC 2	DEC BLDG	HP/2	"MITSUBISHI" MODEL #CMB-P1013NU-GA MAIN BRANCH CIRCUIT CONTROLLER, 13 TOTAL REFRIGERANT BRANCH CIRCUITS, 208V/1ph, COOLING: .0178 KW POWER INPUT, 0.86/0.77 CURRENT INPUT, HEATING: 0.086 POWER INPUT KW, 0.41/0.37 CURRENT INPUT AMPS, 1.08/0.97 MCA, 15 AMPS MOCP. OPERATING WEIGHT: 180 LBS.	8 M5.1	1 M7.3, 1 M7.5	1 2

NOTES: MITSUBISHI VRF SYSTEM AND CONTROLS TO MATCH DRY CREEK UNIFIED SCHOOL DISTRICT CORPORATION YARD STANDARD. NO OTHER MANUFACTURERS TO BE CONSIDERED.

1. CONDENSATE DRAIN PIPING REQUIRED, REFER TO PLUMBING DRAWINGS FOR SIZE AND ROUTING.
2. PROVIDE BC CONTROLLER IN PRE-FAB'D SHEET METAL ENCLOSURE FOR OUTDOOR INSTALLATION.

VRF ENERGY RECOVERY VENTILATOR UNIT SCHEDULE

UNIT	SERVES	ASSOC. OUTDOOR UNIT	DESCRIPTION	MOUNTING DETAIL	CONTROL DIAGRAM	NOTES
ERV 1	DEC BLDG	HP/1 HP/2	"RENEWAIRE" MODEL #HE1.5XRT ENERGY RECOVERY VENTILATOR, 1100 CFM OUTSIDE AIR IN, 1100 CFM EXHAUST AIR OUT, 208V/1ph, SUPPLY AIR HP MOTOR 1.0, FLA 6.2, EXHAUST AIR MOTOR HP 1.0, FLA 6.2, MCA 14 (AMPS), MOPD (AMPS) 15, 475 LBS.	8 M5.1	1 M6.1	1

NOTES:

1. INTERLOCK TO RUN WITH ASSOCIATED HEAT PUMP UNIT VIA FACTORY CONTROLS SYSTEM.

Dry Creek Joint Elementary School District
District Education Center

8849 Cook Riolo Road, Roseville, CA 95747

Dry Creek Joint Elementary School District

FILE NO. 31-15

IDENTIFICATION STAMP DIV. OF THE STATE ARCHITECT OFFICE OF REGULATION SERVICES
APPLICATION NO.
02-115496
AC TM FLS HMR SS BR
DATE 2/23/17

DIVISION OF THE STATE ARCHITECT
revision date: ________ by: ________

REGISTERED PROFESSIONAL ENGINEER
THOMAS A. DUVAL
M 22836
EXPIRES 9/30/18
MECHANICAL
STATE OF CALIFORNIA

DATE SIGNED: 2/15/2017

CAPITAL ENGINEERING CONSULTANTS, INC.
RANCHO CORDOVA, CALIFORNIA
RL — PM - DESIGN TEAM
160110.00 — PROJECT NO.

CONSULTANT

nacht&lewis
600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtlewis.com
916.329.4000

LICENSED ARCHITECT
BRIAN J. MAYTUM
No. C-26867
RENEWAL DATE: 1/31/19
STATE OF CALIFORNIA

ARCHITECT

INITIAL BOX

NO.	DWG BY	DATE	REVIEWED

REVISIONS

NO.	DESCRIPTION	DATE	REV'D
	100% design developement	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017

JOB NO. Y1538.00

SHEET TITLE

Mechanical Schedules

SHEET NO.

M0.3

SHEET ____ OF ____ TOTAL

QC INI

ONE INCH = TWENTY FEET
ONE-SIXTEENTH INCH = ONE FOOT
ONE-EIGHTH INCH = ONE FOOT
ONE-QUARTER INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE INCH = ONE FOOT
ONE AND ONE-HALF INCH = ONE FOOT

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Compliance Scope:	NewEnvelopeAndLighting	Input File Name:	Dry Creek District Ed Center.cibdx

A. PROJECT GENERAL INFORMATION

1.	Project Location (city)	Roseville	8.	Standards Version	Compliance2015
2.	CA Zip Code	95747	9.	Compliance Software (version)	EnergyPro 6.8
3.	Climate Zone	11	10.	Building Orientation (deg)	(N) 0 deg
4.	Total Conditioned Floor Area in Scope	15,147 ft^2	11.	Permitted Scope of Work	NewEnvelopeAndLighting
5.	Total Unconditioned Floor Area	0 ft^2	12.	Building Type(s)	Nonresidential
6.	Total # of Stories (Habitable Above Grade)	1	13	Gas Type	NaturalGas
7.	Total # of dwelling units	0			

B. COMPLIANCE RESULTS FOR PERFORMANCE COMPONENTS (Annual TDV Energy Use, kBtu/ft ^{2}-yr) § 140.1

BUILDING COMPLIES

1. Energy Component	2. Standard Design (TDV)	3. Proposed Design (TDV)	4. Compliance Margin (TDV)	5. Percent Better than Standard
Space Heating	14.4	14.0	0.4	2.8%
Space Cooling	82.2	86.8	-4.6	-5.6%
Indoor Fans	21.4	23.4	-2.0	-9.3%
Heat Rejection	--	--	--	--
Pumps & Misc.	0.9	0.9	--	0.0%
Domestic Hot Water	--	--	--	--
Indoor Lighting	47.4	40.4	7.0	14.8%
COMPLIANCE TOTAL	**166.3**	**165.5**	**0.8**	**0.5%**
Receptacle	106.6	106.6	0.0	0.0%
Process	0.1	0.1	0.0	0.0%
Other Ltg	--	--	--	--
TOTAL	**273.0**	**272.2**	**0.8**	**0.3%**

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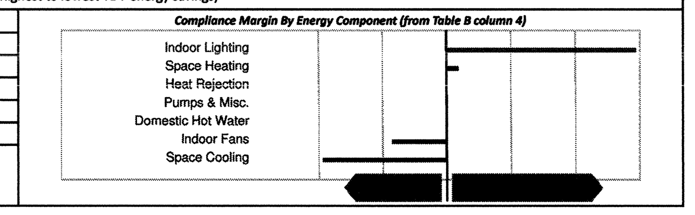
C. PRIORITY PLAN CHECK/ INSPECTION ITEMS (in order of highest to lowest TDV energy savings)

1st	Indoor Lighting: Check lighting
2nd	Space Heating: Check envelope and mechanical
3rd	Heat Rejection: Check envelope and mechanical
4th	Pumps & Misc.: Check mechanical
5th	Domestic Hot Water: Check mechanical
6th	Indoor Fans: Check envelope and mechanical
7th	Space Cooling: Check envelope and mechanical

Compliance Margin By Energy Component (from Table B column 4)



D. EXCEPTIONAL CONDITIONS

The aged solar reflectance and aged thermal emittance must be listed in the Cool Roof Rating Council database of certified products. For projects where initial reflectance is used, the initial reflectance must be listed, and the aged reflectance is calculated by the software program and used in the compliance model.

The building does not include service water heating. Verify that service water heating is not required and is not included in the design.

E. HERS VERIFICATION

This Section Does Not Apply

F. ADDITIONAL REMARKS

None
Provided

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G. COMPLIANCE PATH & CERTIFICATE OF COMPLIANCE SUMMARY

Identify which building components use the performance or prescriptive path for compliance. "NA"= not in project

For components that utilize the performance path, indicate the sheet number that includes mandatory notes on plans.

Building Component	Compliance Path	Compliance Forms *(required for submittal)*	Location of Mandatory Notes on Plans
Envelope	☒ Performance	NRCC-PRF-ENV-DETAILS (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-ENV-01 / 02 / 03 / 04 / 05 / 06-E	
	☐ NA		
Mechanical	☐ Performance	NRCC-PRF-MCH-DETAILS (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-MCH-01 / 02 / 03 / 04 / 05 / 06 / 07-E	
	☒ NA		
Domestic Hot Water	☐ Performance	NRCC-PRF-PLB-DETAILS (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-PLB-01-E	
	☒ NA		
Lighting (Indoor Conditioned)	☒ Performance	NRCC-PRF-LTI-DETAILS (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-LTI-01 / 02 / 03 / 04 / 05-E	
	☐ NA		
Covered Process: Commercial Kitchens	☐ Performance	S2 (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-PRC-01/ 03-E	
	☒ NA		
Covered Process: Computer Rooms	☒ Performance	S3 (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-PRC-01/ 04-E	
	☐ NA		
Covered Process: Laboratory Exhaust	☐ Performance	S4 (section of the NRCC-PRF-01-E)	
	☐ Prescriptive	NRCC-PRC-01/ 09-E	
	☒ NA		

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G. COMPLIANCE PATH & CERTIFICATE OF COMPLIANCE SUMMARY

The following building components are only eligible for prescriptive compliance. Indicate which are relevant to the project.				*The following building components may have mandatory requirements per Part 6. Indicate which are relevant to the project.*			
Yes	NA	Prescriptive Requirement	Compliance Forms	Yes	NA	Mandatory Requirement	Compliance Forms
☐	☒	Lighting (Indoor Unconditioned) §140.6	NRCC-LTI-01 / 02 / 03 / 04 / 05-E			Commissioning: §120.8	
				☐	☒	Simple Systems	NRCC-CXR-01 / 02 / 03 / 05-E
				☐	☒	Complex Systems	NRCC-CXR-01 / 02 / 04 / 05-E
☐	☒	Lighting (Outdoor) §140.7	NRCC-LTO-01 / 02 / 03-E	☐	☒	Electrical: §130.5	NRCC-ELC-01-E
☐	☒	Lighting (Sign) §140.8	NRCC-LTS-01-E	☐	☒	Solar Ready: §110.10	NRCC-SRA-01 / 02-E
☐	☒	Solar Thermal Water Heating: §140.5	NRCC-STH-01-E			Covered Process: §120.6	NRCC-PRC-01-E
				☐	☒	Parking Garage	NRCC-PRC-02-E
				☐	☒	Commercial Refrigeration	NRCC-PRC-05-E
				☐	☒	Warehouse Refrigeration	NRCC-PRC-06/07/08-E
				☐	☒	Compressed Air	NRCC-PRC-10-E
				☐	☒	Process Boilers	NRCC-PRC-11-E

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H. CERTIFICATE OF INSTALLATION, CERTIFICATE OF ACCEPTANCE & CERTIFICATE OF VERIFICATION SUMMARY (NRCI/NRCA/NRCV) –
Documentation Author to indicate which Certificates must be submitted for the features to be recognized for compliance
(Retain copies and verify forms are completed and signed to post in field for Field Inspector to verify).
See Tables G. and H. in MCH and LTI Details Sections for Acceptance Tests and forms by equipment.

Building Component	Compliance Forms *(required for submittal)*	Confirmed: Pass	Confirmed: Fail
Envelope	☒ NRCI-ENV-01-E - For all buildings	☐	☐
	☒ NRCA-ENV-02-F- NFRC label verification for fenestration	☐	☐
Mechanical	☐ NRCI-MCH-01-E - For all buildings with Mechanical Systems	☐	☐
	☐ NRCA-MCH-02-A- Outdoor Air	☐	☐
	☐ NRCA-MCH-03-A – Constant Volume Single Zone HVAC	☐	☐
	☐ NRCA-MCH-04-H- Air Distribution Duct Leakage	☐	☐
	☐ NRCA-MCH-05-A- Air Economizer Controls	☐	☐
	☐ NRCA-MCH-06-A- Demand Control Ventilation	☐	☐
	☐ NRCA-MCH-07-A – Supply Fan Variable Flow Controls	☐	☐
	☐ NRCA-MCH-08-A- Valve Leakage Test	☐	☐
	☐ NRCA-MCH-09-A – Supply Water Temp Reset Controls	☐	☐
	☐ NRCA-MCH-10-A- Hydronic System Variable Flow Controls	☐	☐
	☐ NRCA-MCH-11-A – Auto Demand Shed Controls	☐	☐
	☐ NRCA-MCH-12-A- Packaged Direct Expansion Units	☐	☐
	☐ NRCA-MCH-13-A- Air Handling Units and Zone Terminal Units	☐	☐
	☐ NRCA-MCH-14-A- Distributed Energy Storage	☐	☐
	☐ NRCA-MCH-15-A – Thermal Energy Storage	☐	☐
	☐ NRCA-MCH-16-A- Supply Air Temp Reset Controls	☐	☐
	☐ NRCA-MCH-17-A – Condensate Water Temp Reset Controls	☐	☐
	☐ NRCA-MCH-18-A- Energy Management Controls Systems	☐	☐
	☐ NRCV-MCH-04-H- Duct Leakage Test	☐	☐

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H. CERTIFICATE OF INSTALLATION, CERTIFICATE OF ACCEPTANCE & CERTIFICATE OF VERIFICATION SUMMARY (NRCI/NRCA/NRCV) –
Documentation Author to indicate which Certificates must be submitted for the features to be recognized for compliance
(Retain copies and verify forms are completed and signed to post in field for Field Inspector to verify).
See Tables G. and H. in MCH and LTI Details Sections for Acceptance Tests and forms by equipment.

Building Component	Compliance Forms *(required for submittal)*	Confirmed: Pass	Confirmed: Fail
Plumbing	☐ NRCI-PLB-01-E - For all buildings with Plumbing Systems	☐	☐
	☐ NRCI-PLB-02-E - required on central systems in high-rise residential, hotel/motel application.	☐	☐
	☐ NRCI-PLB-03-E - Single dwelling unit systems in high-rise residential, hotel/motel application.	☐	☐
	☐ NRCI-PLB-21-E - HERS verified central systems in high-rise residential, hotel/motel application.	☐	☐
	☐ NRCI-PLB-22-E - HERS verified single dwelling unit systems in high-rise residential, hotel/motel application.	☐	☐
	☐ NRCV-PLB-21-H- HERS verified central systems in high-rise residential, hotel/motel application.	☐	☐
	☐ NRCV-PLB-22-H - HERS verified single dwelling unit systems in high-rise residential, hotel/motel application.	☐	☐
	☐ NRCI-STH-01-E - Any solar water heating	☐	☐
Indoor Lighting	☒ NRCI-LTI-01-E - For all buildings	☐	☐
	☐ NRCI-LTI-02-E - Lighting control system, or for an Energy Management Control System (EMCS)	☐	☐
	☐ NRCI-LTI-03-E - Line-voltage track lighting integral current limiter, or for a supplementary overcurrent protection panel used to energize only line-voltage track lighting	☐	☐
	☐ NRCI-LTI-04-E - Two interlocked systems serving an auditorium, a convention center, a conference room, or a theater	☐	☐
	☐ NRCI-LTI-05-E - Lighting Control Credit Power Adjustment Factor (PAF)	☐	☐
	☐ NRCI-LTI-06-E - Additional wattage installed in a video conferencing studio	☐	☐
	☐ NRCA-LTI-02-A - Occupancy sensors and automatic time switch controls.	☐	☐
	☐ NRCA-LTI-03-A - Automatic daylighting controls	☐	☐
	☐ NRCA-LTI-04-A - Demand responsive lighting controls	☐	☐
Outdoor Lighting	☐ NRCI-LTO-01-E – Outdoor Lighting	☐	☐
	☐ NRCI-LTO-02-E- EMCS Lighting Control System	☐	☐
	☐ NRCA-LTO-02-A - Outdoor Lighting Control	☐	☐
Sign Lighting	☐ NRCI-LTS-01-E – Sign Lighting	☐	☐
Electrical	☐ NRCI-ELC-01-E - Electrical Power Distribution	☐	☐
Photovoltaic	☐ NRCI-SPV-01-E Photovoltaic Systems	☐	☐

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H. CERTIFICATE OF INSTALLATION, CERTIFICATE OF ACCEPTANCE & CERTIFICATE OF VERIFICATION SUMMARY (NRCI/NRCA/NRCV) –
Documentation Author to indicate which Certificates must be submitted for the features to be recognized for compliance
(Retain copies and verify forms are completed and signed to post in field for Field Inspector to verify).
See Tables G. and H. in MCH and LTI Details Sections for Acceptance Tests and forms by equipment.

Building Component	Compliance Forms *(required for submittal)*	Confirmed: Pass	Confirmed: Fail
Covered Process	☐ NRCI-PRC-01-E Refrigerated Warehouse	☐	☐
	☐ NRCA-PRC-01-F- Compressed Air Systems	☐	☐
	☐ NRCA-PRC-02-F- Kitchen Exhaust	☐	☐
	☐ NRCA-PRC-03-F- Garage Exhaust	☐	☐
	☐ NRCA-PRC-04-F- Refrigerated Warehouse- Evaporator Fan Motor Controls	☐	☐
	☐ NRCA-PRC-05-F- Refrigerated Warehouse- Evaporative Condenser Controls	☐	☐
	☐ NRCA-PRC-06-F- Refrigerated Warehouse- Air Cooled Condenser Controls	☐	☐
	☐ NRCA-PRC-07F- Refrigerated Warehouse- Variable Speed Compressor	☐	☐
	☐ NRCA-PRC-08-F- Electrical Resistance Underslab Heating System	☐	☐

I. ENVELOPE GENERAL INFORMATION (See NRCC-PRF-ENV-DETAILS for more information)

1.	Total Conditioned Floor Area	15,147 ft^2	5.	Number of Floors Above Grade	1
2.	Total Unconditioned Floor Area	0 ft^2	6.	Number of Floors Below Grade	0
3.	Addition Conditioned Floor Area	0 ft^2			
4.	Addition Unconditioned Floor Area	0 ft^2			

7. Opaque Surfaces & Orientation	8. Total Gross Surface Area	9. Total Fenestration Area	10. Window to Wall Ratio	Confirmed: Pass	Confirmed: Fail
North Wall	1,619 ft^2	128 ft^2	07.9%	☐	☐
East Wall	2,842 ft^2	224 ft^2	07.9%	☐	☐
South Wall	1,038 ft^2	64 ft^2	06.2%	☐	☐
West Wall	2,145 ft^2	240 ft^2	11.2%	☐	☐
Total	**7,643 ft^2**	**656 ft^2**	**08.6%**	☐	☐
Roof	16,373 ft^2	155 ft^2	00.9%	☐	☐

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J. FENESTRATION ASSEMBLY SUMMARY § 110.6

1. Fenestration Assembly Name / Tag or I.D.	2. Fenestration Type	3. Certification Method[1]	4. Assembly Method	5. Area ft^2	6. Overall U-factor	7. Overall SHGC	8. Overall VT	9. Status[2]	Confirmed: Pass	Confirmed: Fail
Montecito 8220M w SunCoat Low-E	VerticalFenestration	NFRCRated	SiteBuilt	656	0.34	0.27	0.50	N	☐	☐
Double Metal Clear - Skylight	Skylight	NFRCRated	SiteBuilt	155	0.51	0.73	0.50	N	☐	☐

[1] *Newly installed fenestration shall have a certified NFRC Label Certificate or use the CEC default tables found in Table 110.6-A and Table 110.6-B. Center of Glass (COG) values are for the glass-only, determined by the manufacturer, and are shown for ease of verification. Site-built fenestration values are calculated per Nonresidential Appendix NA6 and are used in the analysis.*
[2] *Status: N - New, A – Altered, E – Existing*

Taking compliance credit for fenestration shading devices? (if "Yes", see NRCC-PRF-ENV-DETAILS for more information)	No

K. OPAQUE SURFACE ASSEMBLY SUMMARY § 120.7/ § 140.3

1. Surface Name	2. Surface Type	3. Area (ft^2)	4. Framing Type	5. Cavity R-Value	6. Continuous R-Value	7. U-Factor / F-Factor / C-Factor	8. Status[1]	Confirmed: Pass	Confirmed: Fail
R-25 Wall w/ 1/2 Polyiso4	ExteriorWall	7643	Wood	25	3	U-Factor: 0.051	N	☐	☐
Slab On Grade7	UndergroundFloor	15147	NA	0	NA	F-Factor: 0.730	N	☐	☐
Wood Framed - R-249	Roof	13675	Metal	24	NA	U-Factor: 0.040	N	☐	☐
R-25 Roof Cathedral41	Roof	2697	Wood	25	NA	U-Factor: 0.038	N	☐	☐
R-19 Wall137	InteriorWall	100	Wood	19	NA	U-Factor: 0.069	N	☐	☐

[1] *Status: N - New, A – Altered, E – Existing*

L. ROOFING PRODUCT SUMMARY § 140.3

1. Product Type	2. Product Density (lb/ft^2)	3. Aged Solar Reflectance	4. Thermal Emittance	5. SRI	6. Cool Roof Credit	7. Roofing Product Description	Confirmed: Pass	Confirmed: Fail
Wood Framed - R-249	4.64771	0.20	0.75	Not Provided	Yes	CRRC Prod. ID: Wood Framed - R-24	☐	☐
R-25 Roof Cathedral41	4.65104	0.63	0.85	Not Provided	Yes	CRRC Prod. ID: R-25 Roof Cathedral	☐	☐

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M. HVAC SYSTEM SUMMARY (see NRCC-PRF-MCH-DETAILS for more information)

This Section Does Not Apply

N. ECONOMIZER & FAN SYSTEMS SUMMARY[1] § 140.4

This Section Does Not Apply

[1] *Mechanical ventilation calculations and exhaust fans are included in the NRCC-PRF-MCH-DETAILS section*

O. EQUIPMENT CONTROLS § 120.2

This Section Does Not Apply

P. SYSTEM DISTRIBUTION SUMMARY § 120.4/ § 140.4(l)

This Section Does Not Apply

Multifamily or Hotel/ Motel Occupancy? (If "Yes", see NRCC-PRF-MCH-DETAILS for DHW system information)	No

Q. INDOOR CONDITIONED LIGHTING GENERAL INFO (see NRCC-PRF-LTI-DETAILS for more info)[3] § 140.6

1. Occupancy Type[1]	2. Conditioned Floor Area[2] (ft^2)	3. Installed Lighting Power (Watts)	4. Lighting Control Credits (Watts)	5. Additional (Custom) Allowance: Area Category Footnotes (Watts)	5. Additional (Custom) Allowance: Tailored Method (Watts)	Confirmed: Pass	Confirmed: Fail
						☐	☐
Office (Greater than 250 square feet in floor area)	10,681	7,583	0	0	0	☐	☐
Corridors, Restrooms, Stairs, and Support Areas	1,022	726	0	0	0	☐	☐
Office (250 square feet in floor area or less)	1,983	1,409	0	0	0	☐	☐
Convention, Conference, Multipurpose and Meeting Center Areas	1,421	1,007	0	0	0	☐	☐
Computer Room	44	31	0	0	0	☐	☐

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Q. INDOOR CONDITIONED LIGHTING GENERAL INFO (see NRCC-PRF-LTI-DETAILS for more info)[3] § 140.6

1. Occupancy Type[1]	2. Conditioned Floor Area[2] (ft^2)	3. Installed Lighting Power (Watts)	4. Lighting Control Credits (Watts)	5. Additional (Custom) Allowance: Area Category Footnotes (Watts)	5. Additional (Custom) Allowance: Tailored Method (Watts)	Confirmed: Pass	Confirmed: Fail
						☐	☐
Building Totals:	15,151	10,756	0	0	0		

[1] *See Table 140.6-C*
[2] *See NRCC-LTI-01-E for unconditioned spaces*
[3] *Lighting information for existing spaces modeled is not included in the table*

R. INDOOR CONDITIONED LIGHTING SCHEDULE (Adapted from NRCC-LTI-01-E)[1] § 130.0

This Section Does Not Apply

[1] *If lighting power densities were used in the compliance model Building Departments will need to check prescriptive forms for Luminaire Schedule details.*

S1. COVERED PROCESS SUMMARY – ENCLOSED PARKING GARAGES § 140.9

This Section Does Not Apply

S2. COVERED PROCESS SUMMARY – COMMERCIAL KITCHENS § 140.9

This Section Does Not Apply

S3. COVERED PROCESS SUMMARY – COMPUTER ROOMS § 140.9

This Section Does Not Apply

S4. COVERED PROCESS SUMMARY – LABORATORY EXHAUSTS § 140.9

This Section Does Not Apply

T. UNMET LOAD HOURS

This Section Does Not Apply

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U. ENERGY USE SUMMARY

Energy Component	Standard Design Site (MWh)	Proposed Design Site (MWh)	Margin (MWh)	Standard Design Site (MBtu)	Proposed Design Site (MBtu)	Margin (MBtu)
Space Heating	0.0	0.0	--	116.7	113.2	3.5
Space Cooling	33.4	35.3	-1.9	--	--	--
Indoor Fans	11.6	12.3	-0.7	--	--	--
Heat Rejection	--	--	--	--	--	--
Pumps & Misc.	0.7	0.8	-0.1	--	--	--
Domestic Hot Water	--	--	--	--	--	--
Indoor Lighting	28.5	24.3	4.2	--	--	--
COMPLIANCE TOTAL	**74.2**	**72.7**	**1.5**	**116.7**	**113.2**	**3.5**
Receptacle	63.6	63.6	0.0	--	--	--
Process	0.1	0.1	0.0	--	--	--
Other Ltg	--	--	--	--	--	--
TOTAL	**137.9**	**136.4**	**1.5**	**116.7**	**113.2**	**3.5**

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DOCUMENTATION AUTHOR'S DECLARATION STATEMENT § 10-103

I certify that this Certificate of Compliance documentation is accurate and complete.

Documentation Author Name: Aaron Wintersmith	Signature:
Company: Capital Engineering Consultants, Inc.	
Address: 11020 Sun Center Dr	Signature Date: 02-14-2017
City/State/Zip: Rancho Cordova CA 95670	CEA Identification (If applicable):
Phone: 916-851-3500	

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1	I hereby affirm that I am eligible under the provisions of Division 3 of the Business and Professions Code to sign this document as the person responsible for its preparation; and that I am licensed in the State of California as a civil engineer, mechanical engineer, electrical engineer, or I am a licensed architect.
2	I affirm that I am eligible under the provisions of Division 3 of the Business and Professions Code by section 5537.2 or 6737.3 to sign this document as the person responsible for its preparation; and that I am a licensed contractor performing this work.
3	I affirm that I am eligible under Division 3 of the Business and Professions Code to sign this document because it pertains to a structure or type of work described as exempt pursuant to Business and Professions Code Sections 5537, 5538 and 6737.1.

Responsible Envelope Designer Name:	Signature:	
Company: nact&lewis		
Address: 600 Q St. Suite 100	Date Signed: 2/22/17	
City/State/Zip: Sacramento Ca 95611	Declaration Statement Type:	
Phone: 916-329-4000	Title: ARCHITECT OF RECORD	License #: C-26867
Responsible Lighting Designer Name:	Signature:	
Company:		
Address:	Date Signed: 2-15-17	
City/State/Zip:	Declaration Statement Type:	
Phone:	Title: Principal	License #: E14987
Responsible Mechanical Designer Name: - specify -	Signature: NOT IN SCOPE	
Company: Capital Engineering		
Address: 11020 Sun Center Dr #100	Date Signed:	
City/State/Zip: Rancho Cordova CA 95670	Declaration Statement Type:	
Phone: 916-851-3500	Title:	License #:

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Dry Creek Joint Elementary School District
District Education Center
8849 Cook Riolo Road, Roseville, CA 95747
Dry Creek Joint Elementary School District

FILE NO. 31-15
IDENTIFICATION STAMP DIV. OF THE STATE ARCHITECT OFFICE OF REGULATION SERVICES
APPLICATION NO. 02-115496
AC___ FLS___ SS___
DATE ___

DIVISION OF THE STATE ARCHITECT
revision date: ________ by: ________

REGISTERED PROFESSIONAL ENGINEER THOMAS A. DUNAY M 22836 EXPIRES 9/30/18 MECHANICAL STATE OF CALIFORNIA

DATE SIGNED: 2/15/2017

CAPITAL ENGINEERING CONSULTANTS, INC.
RANCHO CORDOVA, CALIFORNIA
RL — PM - DESIGN TEAM
160110.00 — PROJECT NO.

CONSULTANT

nacht&lewis
600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtlewis.com
916.329.4000

LICENSED ARCHITECT BRIAN J. MANTUM No. C-26867 RENEWAL DATE: 1/31/19 STATE OF CALIFORNIA

ARCHITECT

INITIAL BOX

NO.	DWG BY	DATE	REVIEWED

REVISIONS

NO.	DESCRIPTION	DATE	REV'D
	100% design developement	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017
JOB NO. Y1538.00
SHEET TITLE
CA Energy Code Compliance Forms

SHEET NO.
M0.4
SHEET____OF____TOTAL

ONE INCH = TWENTY FEET · ONE-SIXTEENTH INCH = ONE FOOT · ONE-EIGHTH INCH = ONE FOOT · ONE-QUARTER INCH = ONE FOOT · ONE-HALF INCH = ONE FOOT · THREE-QUARTERS INCH = ONE FOOT · ONE INCH = ONE FOOT · ONE AND ONE-HALF INCH = ONE FOOT

QC INI

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C. OPAQUE DOOR SUMMARY							Confirmed	
1.	2.	3.	4.	5.	6.	7.	Pass	Fail
Opaque Door Assembly Name / Tag or I.D.	Door Type	Certification Method	Operation	Area	Overall U-factor	Status[1]		
Wood Door16	WoodGreaterThanOrEqualTo1.75inThickDoor	DefaultPerformance	Swinging	126	0.500	N	☐	☐

[1] *Status: N - New, A – Altered, E – Existing*

NRCC-PRF-MCH-DETAILS -SECTION START-

A. MECHANICAL VENTILATION AND REHEAT *(Adapted from 2013-NRCC-MCH-03-E)*
This Section Does Not Apply

B. ZONAL SYSTEM AND TERMINAL UNIT SUMMARY
This Section Does Not Apply

C. EXHAUST FAN SUMMARY
This Section Does Not Apply

D. DHW EQUIPMENT SUMMARY – *(Adapted from NRCC-PLB-01)*
This Section Does Not Apply

E. MULTI-FAMILY CENTRAL DHW SYSTEM DETAILS
This Section Does Not Apply

F. SOLAR HOT WATER HEATING SUMMARY (Adapted from NRCC-STH-01)
This Section Does Not Apply

G. MECHANICAL HVAC ACCEPTANCE TESTS & FORMS *(Adapted from 2013-NRCC-MCH-01-E)* — § RA4
This Section Does Not Apply

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H. EVAPORATIVE COOLER SUMMARY
This Section Does Not Apply

NRCC-PRF-LTI-DETAILS -SECTION START-

A. INDOOR CONDITIONED LIGHTING CONTROL CREDITS (Adapted from NRCC-LTI-02-E) — § 140.6
This Section Does Not Apply

B. INDOOR CONDITIONED LIGHTING MANDATORY LIGHTING CONTROLS (Adapted from NRCC-LTI-02-E) — § 130.1
This Section Does Not Apply
§130.1(a) = Manual area controls; §130.0(b) = Multi Level; §130.1(c) = Auto Shut-Off; §130.1(d) = Mandatory Daylight; §130.1(e) = Demand Responsive

C. TAILORED METHOD CONDITIONED LIGHTING POWER ALLOWANCE SUMMARY AND CHECKLIST (Adapted from NRCC-LTI-04-E)	§ 140.6
General lighting power (see Table D)	0
General lighting power from special function areas (see Table E)	NA
Additional "use it or lose it" (See Table G)	0
Total watts	0

D. GENERAL LIGHTING POWER (Adapted from NRCC-LTI-04-E) — § 140.6-D
This Section Does Not Apply

E. GENERAL LIGHTING FROM SPECIAL FUNCTION AREAS (Adapted from NRCC-LTI-04-E)							§ 140.6(c) 3H	
Room Number	Primary Function Area	Illuminance Value (LUX)	Room Cavity Ratio (Table G)	Allowed LPD	Floor Area (ft^2)	Allowed Watts	Confirmed Pass	Confirmed Fail
NA	NA	NA	NA	NA	NA	NA	☐	☐

Note: Tailored Method for Special Function Areas is not currently implemented

F. ROOM CAVITY RATIO (Adapted from NRCC-LTI-04-E)							
Rectangular Spaces							
Room Number	Task/Activity Description	Room Length (ft)	Room Width (ft)	Room Cavity Height (ft)	RCR	Confirmed Pass	Confirmed Fail
NA	NA	NA	NA	NA	NA	☐	☐

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Non-Rectangular Spaces
This Section Does Not Apply
Note: All applicable spaces are listed under the Non-Rectangular Spaces table

G. ADDITIONAL "USE IT OR LOSE IT" (Adapted from NRCC-LTI-04-E)					Confirmed	
1.	2.	3.	4.			
Wall Display	Combined Floor Display and Task Lighting	Combined Ornamental and Special Effects Lighting	Very Valuable Merchandise	Allowed Watts	Pass	Fail
0	0	0	0	0	☐	☐

5. Wall Display
This Section Does Not Apply

6. Floor Display and Task Lighting
This Section Does Not Apply

7. Combined Ornamental and Special Effects Lighting
This Section Does Not Apply

8. Very Valuable Merchandise
This Section Does Not Apply

H. INDOOR & OUTDOOR LIGHTING ACCEPTANCE TESTS & FORMS (Adapted from NRCC-LTI-01-E and NRCC-LTO-01-E)						§ 130.4	
Declaration of Required Acceptance Certificates (NRCA) –Acceptance Certificates that must be verified in the field. (Retain copies and verify forms are completed and signed to post in field for Field Inspector to verify).							
Test Description		Indoor			Outdoor	Confirmed	
		NRCA-LTI-02-A	NRCA-LTI-03-A	NRCA-LTI-04-A	NRCA-LTO-02-A	Pass	Fail
Equipment Requiring Testing or Verification	# of units	Occ Sensors / Auto Time Switch	Auto Daylight	Demand Responsive	Outdoor Controls		
Occupant Sensors	0	☐	☐	☐	☐	☐	☐
Automatic Time Switch	0	☐	☐	☐	☐	☐	☐
Automatic Daylighting	0	☐	☐	☐	☐	☐	☐

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H. INDOOR & OUTDOOR LIGHTING ACCEPTANCE TESTS & FORMS (Adapted from NRCC-LTI-01-E and NRCC-LTO-01-E)						§ 130.4	
Declaration of Required Acceptance Certificates (NRCA) –Acceptance Certificates that must be verified in the field. (Retain copies and verify forms are completed and signed to post in field for Field Inspector to verify).							
Test Description		Indoor			Outdoor	Confirmed	
		NRCA-LTI-02-A	NRCA-LTI-03-A	NRCA-LTI-04-A	NRCA-LTO-02-A	Pass	Fail
Equipment Requiring Testing or Verification	# of units	Occ Sensors / Auto Time Switch	Auto Daylight	Demand Responsive	Outdoor Controls		
Demand Responsive	0	☐	☐	☐	☐	☐	☐
Outdoor Controls	0	☐	☐	☐	☐	☐	☐

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ENVELOPE MANDATORY MEASURES: NONRESIDENTIAL — ENV-MM

Project Name	Date
Dry Creek JESD District Ed Center	*10/18/2016*

DESCRIPTION

Building Envelope Measures:

§110.8(a):	Installed insulating material shall have been certified by the manufacturer to comply with the California Quality Standards for insulating material, Title 20 Chapter 4, Article 3.
§110.8(c):	All Insulating Materials shall be installed in compliance with the flame spread rating and smoke density requirements of Sections 2602 and 707 of Title 24, Part 2.
§110.8(f):	The opaque portions of framed demising walls in nonresidential buildings shall have insulation with an installed R-value of no less than R-13 between framing members.
§110.7(a):	All Exterior Joints and openings in the building that are observable sources of air leakage shall be caulked, gasketed, weatherstripped or otherwise sealed.
§110.6(a):	Manufactured fenestration products and exterior doors shall have air infiltration rates not exceeding 0.3 cfm/ft.² of window area, 0.3 cfm/ft.² of door area for residential doors, 0.3 cfm/ft.² of door area for nonresidential single doors (swinging and sliding), and 1.0 cfm/ft.² for nonresidential double doors (swinging).
§110.6(a):	Fenestration U-factor shall be rated in accordance with NFRC 100, or the applicable default U-factor.
§110.6(a) :	Fenestration SHGC shall be rated in accordance with NFRC 200, or NFRC 100 for site-built fenestration, or the applicable default SHGC.
§110.6(b):	Site Constructed Doors, Windows and Skylights shall be caulked between the unit and the building, and shall be weatherstripped (except for unframed glass doors and fire doors).

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NRCC-PRF-ENV-DETAILS -SECTION START-

A. OPAQUE SURFACE ASSEMBLY DETAILS				Confirmed	
1.	2.	3.	4.	Pass	Fail
Surface Name	Surface Type	Description of Assembly Layers	Notes		
R-25 Wall w/ 1/2 Polyiso4	ExteriorWall	Stucco - 7/8 in. Vapor permeable felt - 1/8 in. Wood framed wall, 16in. OC, 5.5in., R-25 Gypsum Board - 1/2 in. Cellular polyisocyanurate (unfaced) - 1/2 in. R2.9		☐	☐
Slab On Grade7	UndergroundFloor	Slab Type = UnheatedSlabOnGrade Insulation Orientation = None Insulation R-Value = R0		☐	☐
Wood Framed - R-249	Roof	Asphalt shingles - 1/4 in. Vapor permeable felt - 1/8 in. Plywood - 1/2 in. Air - Metal Wall Framing - 16 or 24 in. OC Wood framed roof, 24in. OC, 9.25in., R-24 Gypsum Board - 1/2 in.		☐	☐
R-25 Roof Cathedral41	Roof	Asphalt shingles - 1/4 in. Vapor permeable felt - 1/8 in. Plywood - 1/2 in. Air - Cavity - Wall Roof Ceiling - 4 in. or more Wood framed roof, 24in. OC, 9.25in., R-25 Gypsum Board - 1/2 in.		☐	☐
R-19 Wall137	InteriorWall	Stucco - 7/8 in. Vapor permeable felt - 1/8 in. Wood framed wall, 16in. OC, 5.5in., R-19 Gypsum Board - 1/2 in.		☐	☐

B. OVERHANG DETAILS (Adapted from NRCC-ENV-02-E)
This Section Does Not Apply

CA Building Energy Efficiency Standards- 2013 Nonresidential Compliance   Report Version: NRCC-PRF-01-E-08122016-3995   Report Generated at: 2016-10-11 10:56:43

Dry Creek Joint Elementary School District
District Education Center
8849 Cook Riolo Road, Roseville, CA 95747
Dry Creek Joint Elementary School District

FILE NO. 31-15
IDENTIFICATION STAMP DIV. OF THE STATE ARCHITECT OFFICE OF REGULATION SERVICES
APPLICATION NO. 02-115496
AC ___ FLS ___ SS ___
DATE ___

DIVISION OF THE STATE ARCHITECT
revision date: ___ by: ___

REGISTERED PROFESSIONAL ENGINEER — THOMAS A. DUVAL — M 22836 — EXPIRES 9/30/18 — MECHANICAL — STATE OF CALIFORNIA
DATE SIGNED: 2/15/2017

CAPITAL ENGINEERING CONSULTANTS, INC.
RANCHO CORDOVA, CALIFORNIA
RL — PM - DESIGN TEAM   160110.00 — PROJECT NO.

CONSULTANT

nacht&lewis
600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtlewis.com
916.329.4000

LICENSED ARCHITECT — BRIAN J. MAYTUM — No. C-26867 — RENEWAL DATE: 1/31/19 — STATE OF CALIFORNIA

ARCHITECT

INITIAL BOX

NO.	DWG BY	DATE	REVIEWED

REVISIONS

NO.	DESCRIPTION	DATE	REV'D
	100% design developement	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017
JOB NO. Y1538.00
SHEET TITLE
CA Energy Code Compliance Forms
SHEET NO.
M0.5
SHEET ____ OF ____ TOTAL

ONE INCH = TWENTY FEET
ONE–SIXTEENTH INCH = ONE FOOT
ONE–EIGHTH INCH = ONE FOOT
ONE–QUARTER INCH = ONE FOOT
ONE–HALF INCH = ONE FOOT
THREE–QUARTERS INCH = ONE FOOT
ONE INCH = ONE FOOT
ONE AND ONE–HALF INCH = ONE FOOT